

CALCULATION SHEET FOR

Oil/Gas Separator

DOCUMENT NO. P-2202

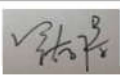
DRAWING NO. 20980145-083 REV.3

This is to certify that these calculations have been made in accordance with the requirement of Section VIII Division 1 of the ASME Boiler and Pressure Vessel Code 2021 Edition and PED2014/68/EU.

Reviewed by AI:

Date:



3			2023.6.17		
2					
1					
0					
Revision	Description of change	Prepared	Date	Approved	Date

WELL-TANKER(ANHUI) M&E EQUIPMENT CO.,LTD.

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Material	PED						ASME	Actual
	t (℃)	Re/t(Mpa)	Re/m(Mpa)	$\frac{2}{3}^* \text{Re/t(Mpa)}$	$\frac{5}{12}^* \text{Re/m(Mpa)}$	$\text{Min}[\frac{2}{3}^* \text{Re/t(Mpa)}, \frac{5}{12}^* \text{Re/m(Mpa)}]$	S(Mpa)	Allowable stress(Mpa)
Q345R(3~16)	0	345	510	230	212.5	212.5	146	146
Q345R(3~16)	150	345	510	230	212.5	212.5	146	146
Q345R(16~36)	0	325	500	216.7	208.3	208.3	143	143
Q345R(16~36)	150	325	500	216.7	208.3	208.3	143	143
SA-105M	0	250	485	166.7	202.1	166.7	138	138
SA-105M	150	250	485	166.7	202.1	166.7	138	138
SA-106B	0	240	415	160	172.9	160	118	118
SA-106B	150	240	415	160	172.9	160	118	118
SA-193B7M(≤100)	0	550	690	366.7	287.5	287.5	138	138
SA-193B7M(≤100)	150	550	690	366.7	287.5	287.5	138	138

DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2019

Analysis Performed by : #Replace this text with your company name and th

Job File : L:\1.WETK\WETK-2021.10.30(DN400-PED-W)\计算书\20

Date of Analysis : Feb 21, 2022 6:12pm

PV Elite 23 SP2, October 2021

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 FileName : 20980145-083, 2, 21
 Vessel Design Summary: Step: 27 6:12pm Feb 21, 2022

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Vessel Design Summary:

ASME Code, Section VIII Division 1, 2019

Diameter Spec : 450.000 mm OD
 Vessel Design Length, Tangent to Tangent 866.00 mm
 Specified Datum Line Distance 25.00 mm
 Internal Design Temperature 150 C
 Internal Design Pressure 1.440 MPa
 External Design Temperature 150 C
 Maximum Allowable Working Pressure 2 MPa
 Shop Test Pressure 2.059 MPa
 Required Minimum Design Metal Temperature 0.0 C
 Warmest Computed Minimum Design Metal Temperature -39.0 C
 Wind Design Code No Wind Loads
 Earthquake Design Code No Seismic

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Normal ized	Impact Tested
Shell	SA/GB 713	Q345R	1/8 < t <= 5/8		No	No
Head	SA/GB 713	Q345R	1/8 < t <= 5/8		No	No
Cover	SA/GB 713	Q345R	5/8 < t <= 1.5		No	No
Nozzle	SA-105	...	...	K03504	No	No
Nozzle	SA-106 B	...	...	K03006	No	No
Nozzle Flg	SA/GB 713	...	5/8 < t <= 1.5		No	No
Nozzle Flg	SA/GB 713	...	5/8 < t <= 1.5		No	No
Hrz Bolting	SA-193 B7	...	<= 2 1/2	G41400	No	No

Normalized is determined based on the UCS-66 material curve selection and Figure UCS-66.
 Impact Tested is based on material selection and material data properties.

Element Pressures and MAWP (MPa & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
L' head	1.440	0.00	2.000	1.0000	No	No
shell	1.440	0.00	2.000	1.0000	N/A	No
baffle plate	1.440	0.00	3.000	1.0000	N/A	No
shell	1.440	0.00	2.000	1.0000	N/A	No
R' - head	1.440	0.00	2.000	1.0000	No	No

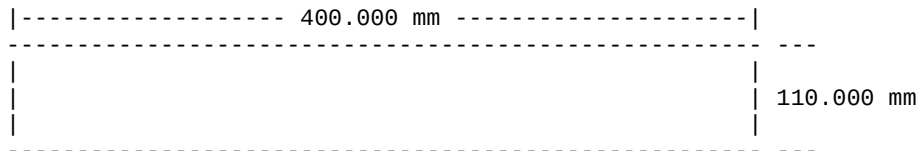
Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	0.0	25.0	6.0	5.0	3.6	3.5	0.85	0.65
Cylinder	706.0	706.0	6.0	6.0	4.2	...	0.70	0.65
Wld Flat	736.0	30.0	30.0	30.0	22.9	...	1.00	1.00
Cylinder	816.0	80.0	6.0	6.0	4.3	...	0.70	0.65
Ellipse	841.0	25.0	6.0	5.0	3.6	3.5	0.85	0.65

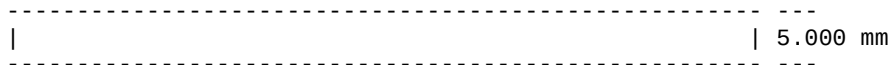
Saddle Parameters:

Saddle Width	110.000	mm
Saddle Bearing Angle	90.000	deg.
Centerline Dimension	330.000	mm
Wear Pad Width	90.000	mm
Wear Pad Thickness	5.000	mm
Wear Pad Bearing Angle	90.000	deg.
Distance from Saddle to Tangent	40.000	mm
Baseplate Length	400.000	mm
Baseplate Thickness	5.000	mm
Baseplate Width	110.000	mm
Number of Ribs (including outside ribs)	2	
Rib Thickness	5.000	mm
Web Thickness	5.000	mm
Height of Center Web	110.000	mm

Baseplate Sketch



Baseplate Plan View



Baseplate Side View

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	1821.48	N
Maximum Transverse Saddle Shear Load	0.00	N
Maximum Longitudinal Saddle Shear Load	0.00	N

Note: The Wind and Seismic loads were not load factored in this analysis.

Summary of Maximum Saddle Loads, Hydrotest Case :

Maximum Vertical Saddle Load	2522.47	N
Maximum Transverse Saddle Shear Load	0.00	N
Maximum Longitudinal Saddle Shear Load	0.00	N

Weights:

Fabricated - Bare W/O Removable Internals	342.1	kg
Shop Test - Fabricated + Water (Full)	498.9	kg
Shipping - Fab. + Rem. Intls.+ Shipping App.	342.1	kg
Erected - Fab. + Rem. Intls.+ Insul. (etc)	342.1	kg
Empty - Fab. + Intls. + Details + Wghts.	342.1	kg
Operating - Empty + Operating Liquid (No CA)	342.1	kg
Field Test - Empty Weight + Water (Full)	498.9	kg

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 Nozzle Summary: Step: 25 6:12pm Feb 21, 2022

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Nozzle Calculation Summary:

Description	MAWP MPa	Ext	MAPNC MPa	UG-45	[tr] mm	Weld Path	Areas or Stresses
Noz i1	2	...	...	OK	3.50	OK	No Calc[*]
Noz a1	2	...	...	OK	3.50	OK	No Calc[*]
Noz b	2	...	...	OK	3.50	OK	No Calc[*]
Noz b	2	...	...	OK	3.50	OK	No Calc[*]
Noz d	2	...	...	OK	3.50	OK	No Calc[*]
Noz d	2	...	...	OK	3.50	OK	No Calc[*]
Noz g1	2	...	...	OK	3.50	OK	No Calc[*]
Noz g1	2	...	...	OK	3.50	OK	No Calc[*]
Noz x-2	2	...	...	OK	3.50	OK	No Calc[*]
Noz x-2	2	...	...	OK	3.50	OK	No Calc[*]
Noz g2	2	...	...	OK	3.50	OK	No Calc[*]
Noz x-1	2	...	...	OK	3.50	OK	No Calc[*]
Noz x-1	2	...	...	OK	3.50	OK	No Calc[*]
Noz c	2	...	...	OK	3.50	OK	No Calc[*]
Noz h	2	...	...	OK	3.50	OK	No Calc[*]
Noz h	2	...	...	OK	3.50	OK	No Calc[*]
Noz a2	2	...	...	OK	3.50	OK	No Calc[*]
Noz f	2	...	...	OK	3.50	OK	No Calc[*]
Noz f	2	...	...	OK	3.50	OK	No Calc[*]
Noz i2	2	...	...	OK	3.50	OK	No Calc[*]

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 2 Nozzle : Noz f
 Minimum MAWP Shells/Flanges : 2 Element : shell
 Minimum MAPnc Shells/Flanges : 3 Element : shell
 Computed Vessel M.A.W.P. : 2 MPa

[*] - This was a small opening and the areas were not computed.

Note: MAWPs (Internal Case) shown above are at the High Point.

Check the Spatial Relationship between the Nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
10	Noz i1	0.000	0.000	128.000
20	Noz a1	431.500	270.000	36.670
20	Noz b	566.500	115.652	44.000
20	Noz d	266.500	118.437	66.000
20	Noz g1	226.500	61.563	40.000
40	Noz x-2	796.000	52.695	68.922
40	Noz g2	796.000	90.000	40.000
40	Noz x-1	796.000	110.262	40.000
40	Noz c	796.000	0.000	62.607
40	Noz h	796.000	141.189	54.000
40	Noz a2	796.000	270.000	36.670
40	Noz f	796.000	-27.036	66.000
50	Noz i2	0.000	0.000	128.000

The nozzle spacing is computed by the following:

= Sqrt($l_l^2 + l_c^2$) where

l_l - Arc length along the inside vessel surface in the long. direction.

l_c - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

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Nozzle Summary:

	Step:	25	6:12pm	Feb 21, 2022
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No interference violations have been detected!

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 FileName : 20980145-083, 2, 21
 Nozzle Schedule: Step: 24 6:12pm Feb 21, 2022

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
Noz a1	0.743 in	80	Non	1.050	3.910	...	...	21.4	...
Noz a2	0.743 in	80	Non	1.050	3.910	...	...	26.4	...
Noz g1	0.984 in Actual	WN	Non	0.984	3.500	...	...	29.4	...
Noz g2	0.984 in Actual	Non	Non	0.984	3.500	...	...	21.3	...
Noz b	1.063 in Actual	Non	Non	1.063	3.500	...	...	38.9	...
Noz x-2	0.957 in	80	Non	1.315	4.550	...	...	56.6	...
Noz x-1	0.957 in	80	Non	1.315	4.550	...	...	56.6	...
Noz h	1.260 in Actual	Non	Non	1.260	3.500	...	...	49.2	...
Noz c	1.378 in Actual	Non	Non	1.378	3.500	...	...	26.6	...
Noz d	1.496 in Actual	Non	Non	1.496	3.500	...	...	38.7	...
Noz f	1.496 in Actual	Non	Non	1.496	3.500	...	...	38.0	...
Noz i1	2.992 in Actual	Non	Non	2.992	6.000	...	...	16.7	...
Noz i2	2.992 in Actual	Non	Non	2.992	6.000	...	...	16.7	...

General Notes for the above table:

The Cut Length is the Outside Projection + Inside Projection + Drop +
 In Plane Shell Thickness. This value does not include weld gaps,
 nor does it account for shrinkage.

In the case of Oblique Nozzles, the Outside Diameter must
 be increased. The Re-Pad WIDTH around the nozzle is calculated as follows:
 Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller
 and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
Noz a1	SA-106 B	5.000	5.000	...	...	...
Noz a2	SA-106 B	5.000	5.000	...	...	...
Noz g1	SA-105	5.000	5.000	...	...	...
Noz g2	SA-105	5.000	5.000	...	...	...
Noz b	SA-105	5.000	5.000	...	...	...
Noz x-2	SA-106 B	5.000	5.000	...	...	...
Noz x-1	SA-106 B	5.000	5.000	...	...	...
Noz h	SA-105	5.000	5.000	...	...	...
Noz c	SA-105	5.000	5.000	...	...	...
Noz d	SA-105	5.000	5.000	...	...	...
Noz f	SA-105	5.000	5.000	...	...	...
Noz i1	SA-105	3.750	6.000	...	...	...
Noz i2	SA-105	3.750	6.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
Noz a1	406.500	270.0	15.00	0.00	shell
Noz a2	771.000	270.0	20.00	0.00	shell
Noz g1	201.500	61.6	15.00	0.00	shell

Noz g2	771.000	90.0	15.00	0.00	shell
Noz b	541.500	115.7	25.00	0.00	shell
Noz x-2	771.000	52.7	30.00	0.00	shell
Noz x-1	771.000	110.3	30.00	0.00	shell
Noz h	771.000	141.2	15.00	0.00	shell
Noz c	771.000	0.0	20.00	0.00	shell
Noz d	241.500	118.4	20.00	0.00	shell
Noz f	771.000	-27.0	20.00	0.00	shell
Noz i1	...	0.0	10.00	0.00	L' head
Noz i2	...	0.0	10.00	0.00	R' head

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QTY	DESCRIPTION	MATERIAL
2	ELLIPTICAL HEAD: 2.0 X 1, 6.0mm THK X 450.0mm ID X 25.0mm	SA/GB 713
1	CYLINDER: 6.0mm THK X 450.0mm ID X 706.0mm	SA/GB 713
1	FLAT HEAD: 30.0mm THK X 462.0mm OD	SA/GB 713
1	CYLINDER: 6.0mm THK X 450.0mm ID X 80.0mm	SA/GB 713
2	SADDLE: 109mm X 90 DEG	
1	NAMEPLATE	...

Minimum Design Metal Temperature Results Summary :

Description	Notes	Curve	Basic MDMT C	Reduced MDMT C	UG-20(f) MDMT C	Thickness ratio	Gov Thk mm	E*	PWHT reqd
L' head	[10]	B	-29	-48	-29	0.608	5.000	0.85	No
L' head	[7]	B	-29	-48	-29	0.492	6.000	0.85	No
shell	[8]	B	-29	-48	-29	0.563	6.000	0.80	No
baffle plate	[12]	B	-29	-39	-29	0.818	7.500	1.00	No
shell	[8]	B	-29	-48	-29	0.575	6.000	0.80	No
R' - head	[10]	B	-29	-48	-29	0.608	5.000	0.85	No
R' - head	[7]	B	-29	-48	-29	0.492	6.000	0.85	No
Noz i1	[1]	A	-8	-40	-29	0.502	5.000	1.00	No
Noz a1	[1]	B	-29	-104	-29	0.028	4.845	1.00	No
Noz b	[1]	A	-8	-104	-29	0.056	3.500	1.00	No
Noz d	[1]	A	-8	-104	-29	0.079	3.500	1.00	No
Noz g1	[1]	A	-8	-104	-29	0.052	3.500	1.00	No
Noz x-2	[1]	B	-29	-104	-29	0.065	4.245	1.00	No
Noz g2	[1]	A	-8	-104	-29	0.052	3.500	1.00	No
Noz x-1	[1]	B	-29	-104	-29	0.065	4.245	1.00	No
Noz c	[1]	A	-8	-104	-29	0.073	3.500	1.00	No
Noz h	[1]	A	-8	-104	-29	0.067	3.500	1.00	No
Noz a2	[1]	B	-29	-104	-29	0.028	4.845	1.00	No
Noz f	[1]	A	-8	-104	-29	0.079	3.500	1.00	No
Noz i2	[1]	A	-8	-40	-29	0.502	5.000	1.00	No
Warmest MDMT:			-8	-39					

Required Minimum Design Metal Temperature 0.0 C
 Warmest Computed Minimum Design Metal Temperature -39.0 C

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).
 The Basic MDMT includes the (30F) PWHT credit if applicable.

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 Warnings and Errors: Step: 0 6:12pm Feb 21, 2022

Class From To : Basic Element Checks.

Note	20	40	There is a high jump in the Joint Eff.
Note	30	50	There is a high jump in the Joint Eff.

Class From To: Check of Additional Element Data

Please insure the C factor entered is in accordance with Figure UG-34.

There were no geometry errors or warnings.

Note:

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

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 Input Echo: Step: 1 6:12pm Feb 21, 2022

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PV Elite Vessel Analysis Program: Input Data

Design Internal Pressure (for Hydrotest)	1.44	MPa
Design Internal Temperature	150.0	C
Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Minimum Design Metal Temperature	0.0	C
Type of Construction	Welded	
Special Service	Air/Water/Steam	
Degree of Radiography	None	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	
Perform a Corroded Hydrotest	N	

Shop Pressure Test:

Type of Pressure Test	1.43 * MAWP (PED)
Pressure Test Position	Vertical

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW

Wind Design Code	No Wind Loads
------------------	---------------

Seismic Design Code	No Seismic
---------------------	------------

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Material Database Year	Current w/Addenda or Code Year
------------------------	--------------------------------

Configuration Directives:

Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No

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Using Metric Material Databases, ASME II D	No
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT calcs. (Div. 1)	Yes

Complete Listing of Vessel Elements and Details:

Element From Node	10
Element To Node	20
Element Type	Elliptical
Description	L' head
Distance "FROM" to "TO"	25 mm
Inside Diameter	450 mm
Element Thickness	5 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	6 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	1.44 MPa
Design Temperature Internal Pressure	150 C
Design External Pressure	0 MPa
Design Temperature External Pressure	150 C
Effective Diameter Multiplier	1.2
Material Name	SA/GB 713
Allowable Stress, Ambient	145.5 MPa
Allowable Stress, Operating	145.5 MPa
Allowable Stress, Hydrotest	310.3 MPa
Material Density	0.00775 kg/cm^3
P Number Thickness	30.988 mm
Yield Stress, Operating	305.3 MPa
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	
Class / Thickness / Grade	Q345R:: 1/8 < t <= 5/8
Product Form	Plate
Efficiency, Longitudinal Seam	0.85
Efficiency, Circumferential Seam	0.65
Elliptical Head Factor	2.0
Weld is pre-Heated	No

Element From Node	10
Detail Type	Nozzle
Detail ID	Noz i1
Dist. from "FROM" Node / Offset dist	0 mm
Nozzle Diameter	2.992 in.
Nozzle Schedule	None
Nozzle Class	None
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0.8012 N
Grade of Attached Flange	None
Nozzle Matl	SA-105

Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	shell
Distance "FROM" to "TO"	706 mm
Inside Diameter	450 mm
Element Thickness	6 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	6 mm
External Corrosion Allowance	0 mm

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Design Internal Pressure	1.44	MPa
Design Temperature Internal Pressure	150	C
Design External Pressure	0	MPa
Design Temperature External Pressure	150	C
Effective Diameter Multiplier	1.2	
Material Name	SA/GB 713	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Circumferential Seam	0.65	
Weld is pre-Heated	No	

Element From Node	20	
Detail Type	Saddle	
Detail ID	L- Sdl	
Dist. from "FROM" Node / Offset dist	40	mm
Width of Saddle	110	mm
Height of Saddle at Bottom	330	mm
Saddle Contact Angle	90.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	90	mm
Thickness of Wear Plate	5	mm
Contact Angle, Wear Plate (degrees)	90.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	20	
Detail Type	Saddle	
Detail ID	Sdl 2 Fr20	
Dist. from "FROM" Node / Offset dist	706	mm
Width of Saddle	110	mm
Height of Saddle at Bottom	330	mm
Saddle Contact Angle	90.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	90	mm
Thickness of Wear Plate	5	mm
Contact Angle, Wear Plate (degrees)	90.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

Element From Node	20	
Detail Type	Nozzle	
Detail ID	Noz a1	
Dist. from "FROM" Node / Offset dist	406.5	mm
Nozzle Diameter	0.75	in.
Nozzle Schedule	160	
Nozzle Class	None	
Layout Angle	270.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0.4362	N
Grade of Attached Flange	None	
Nozzle Matl	SA-106 B	

Element From Node	20	
Detail Type	Nozzle	
Detail ID	Noz b	
Dist. from "FROM" Node / Offset dist	541.5	mm
Nozzle Diameter	1.063	in.
Nozzle Schedule	None	

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Nozzle Class	None	
Layout Angle	115.652	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0.6088	N
Grade of Attached Flange	None	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	Noz d	
Dist. from "FROM" Node / Offset dist	241.5	mm
Nozzle Diameter	1.496	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	118.437	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0.7496	N
Grade of Attached Flange	None	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Nozzle	
Detail ID	Noz g1	
Dist. from "FROM" Node / Offset dist	201.5	mm
Nozzle Diameter	0.984	in.
Nozzle Schedule	None	
Nozzle Class	2500	
Layout Angle	61.563099	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	52.098	N
Grade of Attached Flange	GR 2.1	
Nozzle Matl	SA-105	
Element From Node	20	
Detail Type	Weight	
Detail ID	WGHT:[1 OF 1]	
Dist. from "FROM" Node / Offset dist	80	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	231	mm
Element From Node	20	
Detail Type	Weight	
Detail ID	WGHT:[2 OF 2]	
Dist. from "FROM" Node / Offset dist	406.5	mm
Miscellaneous Weight	980.6	N
Offset from Element Centerline	231	mm

Element From Node	30	
Element To Node	40	
Element Type	Flat	
Description	baffle plate	
Distance "FROM" to "TO"	30	mm
Element Outside Diameter	462	mm
Element Thickness	30	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	30	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.44	MPa
Design Temperature Internal Pressure	150	C
Design External Pressure	0	MPa
Design Temperature External Pressure	150	C
Effective Diameter Multiplier	1.2	
Material Name	SA/GB 713	

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Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Flat Head Attachment Factor	0.2	
Small diameter if Non-Circular	0	mm
Weld is pre-Heated	No	

Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	shell	
Distance "FROM" to "TO"	80	mm
Inside Diameter	450	mm
Element Thickness	6	mm
Internal Corrosion Allowance	1	mm
Nominal Thickness	6	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	1.44	MPa
Design Temperature Internal Pressure	150	C
Design External Pressure	0	MPa
Design Temperature External Pressure	150	C
Effective Diameter Multiplier	1.2	
Material Name	SA/GB 713	
Efficiency, Longitudinal Seam	0.7	
Efficiency, Circumferential Seam	0.65	
Weld is pre-Heated	No	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Noz x-2	
Dist. from "FROM" Node / Offset dist	35	mm
Nozzle Diameter	1.25	in.
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	52.694801	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	N
Grade of Attached Flange	None	
Nozzle Matl	SA-106 B	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Noz g2	
Dist. from "FROM" Node / Offset dist	35	mm
Nozzle Diameter	0.984	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	21.165	N
Grade of Attached Flange	None	
Nozzle Matl	SA-105	

Element From Node	40	
Detail Type	Nozzle	
Detail ID	Noz x-1	
Dist. from "FROM" Node / Offset dist	35	mm
Nozzle Diameter	1.25	in.
Nozzle Schedule	80	
Nozzle Class	150	
Layout Angle	110.262	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	1.556	N
Grade of Attached Flange	None	

Nozzle Matl	SA-106 B
Element From Node	40
Detail Type	Nozzle
Detail ID	Noz c
Dist. from "FROM" Node / Offset dist	35 mm
Nozzle Diameter	1.378 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0.6844 N
Grade of Attached Flange	None
Nozzle Matl	SA-105
Element From Node	40
Detail Type	Nozzle
Detail ID	Noz h
Dist. from "FROM" Node / Offset dist	35 mm
Nozzle Diameter	1.26 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	141.189
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	22.356 N
Grade of Attached Flange	None
Nozzle Matl	SA-105
Element From Node	40
Detail Type	Nozzle
Detail ID	Noz a2
Dist. from "FROM" Node / Offset dist	35 mm
Nozzle Diameter	0.75 in.
Nozzle Schedule	160
Nozzle Class	150
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0.4933 N
Grade of Attached Flange	None
Nozzle Matl	SA-106 B
Element From Node	40
Detail Type	Nozzle
Detail ID	Noz f
Dist. from "FROM" Node / Offset dist	35 mm
Nozzle Diameter	1.496 in.
Nozzle Schedule	None
Nozzle Class	150
Layout Angle	-27.0357
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	0 N
Grade of Attached Flange	None
Nozzle Matl	SA-105

Element From Node	50
Element To Node	60
Element Type	Elliptical
Description	R'- head
Distance "FROM" to "TO"	25 mm
Inside Diameter	450 mm
Element Thickness	5 mm
Internal Corrosion Allowance	1 mm
Nominal Thickness	6 mm

Element From Node	50	
Detail Type	Nozzle	
Detail ID	Noz i2	
Dist. from "FROM" Node / Offset dist	0	mm
Nozzle Diameter	2.992	in.
Nozzle Schedule	None	
Nozzle Class	150	
Layout Angle	0.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0.8012	N
Grade of Attached Flange	None	
Nozzle Matl	SA-105	

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
L' head		25	...	25	...
shell		731	...	706	...
baffle plate		761	...	30	...
shell		841	...	80	...
R' - head		866	...	25	...

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd MPa	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) MPa
L' head		1.44	6	1	450	123.68
shell		1.44	6	1	450	101.85
baffle plate		1.44	30	1	462	142.7
shell		1.44	6	1	450	99.89
R' - head		1.44	6	1	450	123.68

Element Required Thickness and MAWP :

From	To	Design Pressure MPa	M.A.W.P. Corroded MPa	M.A.P. New & Cold MPa	Minimum Thickness mm	Required Thickness mm
L' head		1.44	2.19793	2.74224	5	3.61904
shell		1.44	2.2238	2.67323	6	4.22263
baffle plate		1.44	2.51385	2.6902	30	22.9488
shell		1.44	2.181	2.62178	6	4.28641
R' - head		1.44	2.19793	2.74224	5	3.61904
Minimum			2	3		

MAWP: 2 MPa, limited by: Nozzle Reinforcement.

Elements Suitable for Design Internal Pressure.

Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2019

Elliptical Head From 10 To 20 SA/GB 713 , UCS-66 Crv. B at 150 C

L' head

Required Thickness due to Internal Pressure [tr]:
 $= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P)$ Appendix 1-4 (c)
 $= (1.44 \cdot 452.0 \cdot 0.994) / (2 \cdot 145.5 \cdot 0.85 - 0.2 \cdot 1.44)$
 $= 2.6190 + 1.0000 = 3.6190 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
 $= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t)$ per Appendix 1-4 (c)
 $= (2 \cdot 145.5 \cdot 0.85 \cdot 4.0) / (0.994 \cdot 452.0 + 0.2 \cdot 4.0)$
 $= 2 \text{ MPa}$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t)$ per Appendix 1-4 (c)
 $= (2 \cdot 145.5 \cdot 0.85 \cdot 5.0) / (1.0 \cdot 450.0 + 0.2 \cdot 5.0)$
 $= 3 \text{ MPa}$

Actual stress at given pressure and thickness, corroded [Sact]:
 $= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$
 $= (1.44 \cdot (0.994 \cdot 452.0 + 0.2 \cdot 4.0)) / (2 \cdot 0.85 \cdot 4.0)$
 $= 95.326 \text{ MPa}$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)

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$$= (1.44 * 226.0) / (145.5 * 0.85 - 0.6 * 1.44) + 1.0$$

$$= 3.650 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

$$= (S * E * t) / (R + 0.6 * t) \text{ per UG-27 (c)(1)}$$

$$= (145.5 * 0.85 * 5.0) / (226.0 + 0.6 * 5.0)$$

$$= 3 \text{ MPa}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 * \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (452.0 / (2 * 113.5))^2) / 6$$

$$= 0.994139$$

This material's thickness seems to be in range per the material specification.

Percent Elong. per UCS-79, VIII-1-01-57 $(75 * t_{nom} / R_f) * (1 - R_f / R_o)$ 5.660 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 5.0, tr = 2.86, c = 1.0 mm, E* = 0.85
 Thickness Ratio = $tr * (E^*) / (tg - c) = 0.608$, Temp. Reduction = 22 C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 6.0, tr = 2.895, c = 1.0 mm, E* = 0.85
 Thickness Ratio = $tr * (E^*) / (tg - c) = 0.492$, Temp. Reduction = 33 C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 C

Cylindrical Shell From 20 To 30 SA/GB 713, UCS-66 Crv. B at 150 C

shell

Required Thickness due to Internal Pressure [tr]:

$$= (P * R) / (S * E - 0.6 * P) \text{ per UG-27 (c)(1)}$$

$$= (1.44 * 226.0) / (145.5 * 0.7 - 0.6 * 1.44)$$

$$= 3.2226 + 1.0000 = 4.2226 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S * E * t) / (R + 0.6 * t) \text{ per UG-27 (c)(1)}$$

$$= (145.5 * 0.7 * 5.0) / (226.0 + 0.6 * 5.0)$$

$$= 2 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S * E * t) / (R + 0.6 * t) \text{ per UG-27 (c)(1)}$$

$$= (145.5 * 0.7 * 6.0) / (225.0 + 0.6 * 6.0)$$

$$= 3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P * (R + 0.6 * t)) / (E * t)$$

$$= (1.44 * (226.0 + 0.6 * 5.0)) / (0.7 * 5.0)$$

$$= 94.217 \text{ MPa}$$

This material's thickness seems to be in range per the material specification.

% Elongation per Table UG-79-1 $(50 * t_{nom} / R_f * (1 - R_f / R_o))$ 1.316 %

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 6.0, tr = 3.521, c = 1.0 mm, E* = 0.8

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Thickness Ratio = $tr * (E^*) / (tg - c) = 0.563$, Temp. Reduction = 26 C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 C
 Min Metal Temp. at Required thickness (UCS 66.1) -48 C

Welded Flat Head From 30 To 40 SA/GB 713 , UCS-66 Crv. B at 150 C

baffle plate

Longitudinal Joint: Full Radiography per UW-11(a) Type 1
 Circumferential Joint: Full Radiography per UW-11(a) Type 1

Required Thickness due to Internal Pressure [tr]:

$$= d * \sqrt{Z * C * P / (S * E)} \text{ per UG-34 (c)(3)}$$

$$= 450.0 * \sqrt{1.0 * 0.236 * 1.44 / (142.7 * 1.0)}$$

$$= 21.9488 + 1.0000 = 22.9488 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (t/d)^2 * ((S * E) / (C * Z)) \text{ per UG-34 (c)(3)}$$

$$= ((29.0) / 450.0)^2 * ((142.7 * 1.0) / (0.236 * 1.0))$$

$$= 3 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (t/d)^2 * ((S * E) / (C * Z)) \text{ per UG-34 (c)(3)}$$

$$= (30.0 / 450.0)^2 * ((142.7 * 1.0) / (0.236 * 1.0))$$

$$= 3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (Z * C * P) / (((t/d)^2) * E)$$

$$= (1.0 * 0.236 * 1.44) / (((29.0) / 450.0)^2 * 1.0)$$

$$= 81.743 \text{ MPa}$$

This material's thickness seems to be in range per the material specification.

Minimum Design Metal Temperature Results:

tg = 7.5, tg_sr = 30.0, tr = 23.734, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr * (E^*) / (tg_sr - c) = 0.818$, Temp. Reduction = 10 C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 C
 Min Metal Temp. at Required thickness (UCS 66.1) -39 C

Cylindrical Shell From 40 To 50 SA/GB 713 , UCS-66 Crv. B at 150 C

shell

Required Thickness due to Internal Pressure [tr]:

$$= (P * R) / (S * E - 0.6 * P) \text{ per UG-27 (c)(1)}$$

$$= (1.44 * 226.0) / (142.7 * 0.7 - 0.6 * 1.44)$$

$$= 3.2864 + 1.0000 = 4.2864 \text{ mm}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

$$= (S * E * t) / (R + 0.6 * t) \text{ per UG-27 (c)(1)}$$

$$= (142.7 * 0.7 * 5.0) / (226.0 + 0.6 * 5.0)$$

$$= 2 \text{ MPa}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S * E * t) / (R + 0.6 * t) \text{ per UG-27 (c)(1)}$$

$$= (142.7 * 0.7 * 6.0) / (225.0 + 0.6 * 6.0)$$

$$= 3 \text{ MPa}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P * (R + 0.6 * t)) / (E * t)$$

$$= (1.44 * (226.0 + 0.6 * 5.0)) / (0.7 * 5.0)$$

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$$= 94.217 \text{ MPa}$$

This material's thickness seems to be in range per the material specification.

% Elongation per Table UG-79-1 ($50 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 1.316 %

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 6.0$, $t_r = 3.591$, $c = 1.0$ mm, $E^* = 0.8$
 Thickness Ratio = $t_r \cdot (E^*) / (t_g - c) = 0.575$, Temp. Reduction = 25 C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 C

Elliptical Head From 50 To 60 SA/GB 713 , UCS-66 Crv. B at 150 C

R'- head

Required Thickness due to Internal Pressure [t_r]:
 $= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P)$ Appendix 1-4 (c)
 $= (1.44 \cdot 452.0 \cdot 0.994) / (2 \cdot 145.5 \cdot 0.85 - 0.2 \cdot 1.44)$
 $= 2.6190 + 1.0000 = 3.6190 \text{ mm}$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:
 $= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t)$ per Appendix 1-4 (c)
 $= (2 \cdot 145.5 \cdot 0.85 \cdot 4.0) / (0.994 \cdot 452.0 + 0.2 \cdot 4.0)$
 $= 2 \text{ MPa}$

Maximum Allowable Pressure, New and Cold [MAPNC]:
 $= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t)$ per Appendix 1-4 (c)
 $= (2 \cdot 145.5 \cdot 0.85 \cdot 5.0) / (1.0 \cdot 450.0 + 0.2 \cdot 5.0)$
 $= 3 \text{ MPa}$

Actual stress at given pressure and thickness, corroded [S_{act}]:
 $= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$
 $= (1.44 \cdot (0.994 \cdot 452.0 + 0.2 \cdot 4.0)) / (2 \cdot 0.85 \cdot 4.0)$
 $= 95.326 \text{ MPa}$

Straight Flange Required Thickness:
 $= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c$ per UG-27 (c)(1)
 $= (1.44 \cdot 226.0) / (145.5 \cdot 0.85 - 0.6 \cdot 1.44) + 1.0$
 $= 3.650 \text{ mm}$

Straight Flange Maximum Allowable Working Pressure:
 $= (S \cdot E \cdot t) / (R + 0.6 \cdot t)$ per UG-27 (c)(1)
 $= (145.5 \cdot 0.85 \cdot 5.0) / (226.0 + 0.6 \cdot 5.0)$
 $= 3 \text{ MPa}$

Factor K, corroded condition [K_{cor}]:
 $= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$
 $= (2 + (452.0 / (2 \cdot 113.5))^2) / 6$
 $= 0.994139$

This material's thickness seems to be in range per the material specification.

Percent Elong. per UCS-79, VIII-1-01-57 ($75 \cdot t_{nom} / R_f \cdot (1 - R_f / R_o)$) 5.660 %
 Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, $t_g = 5.0$, $t_r = 2.86$, $c = 1.0$ mm, $E^* = 0.85$
 Thickness Ratio = $t_r \cdot (E^*) / (t_g - c) = 0.608$, Temp. Reduction = 22 C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 C

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MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 6.0, tr = 2.895, c = 1.0 mm, E* = 0.85
 Thickness Ratio = $tr \cdot (E^*) / (tg - c) = 0.492$, Temp. Reduction = 33 C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 C
Min Metal Temp. at Required thickness (UCS 66.1)	-48 C

Note: Heads and Shells Exempted to -20F (-29C) by paragraph UG-20F

Hydrostatic Test Pressure Results:

Pressure per UG99b	= 1.30 * M.A.W.P. * Sa/S	2.044 MPa
Pressure per UG99b[35]	= 1.30 * Design Pres * Sa/S	1.872 MPa
Pressure per UG99c	= 1.30 * M.A.P. - Head(Hyd)	3.404 MPa
Pressure per UG100	= 1.10 * M.A.W.P. * Sa/S	1.729 MPa
Pressure per PED	= max(1.43*DP, 1.25*DP*ratio)	2.059 MPa
Pressure per App 27-4	= M.A.W.P.	1.572 MPa

Pressure per the PED:

= max(1.43 * Design Pressure, 1.25 * Design Pressure * Stress ratio)
 = max(1.43 * 1.44, 1.25 * 1.44 * 1.0)
 = max(2.059, 1.8)
 = 2.059 MPa

Vertical Test performed per: PED

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (MPa):

From To	Stress	Allowable	Ratio	Pressure
L' head	109.5	310.3	0.353	2.06
shell	112.3	310.3	0.362	2.06
baffle plate	146.3	291.7	0.501	2.06
shell	112.3	291.7	0.385	2.06
R' - head	109.5	310.3	0.353	2.06

Stress ratios for Nozzle and Pad Materials (MPa):

Description	Pad/Nozzle	Ambient	Operating	Ratio
Noz i1	Nozzle	137.90	137.90	1.000
Noz a1	Nozzle	117.90	117.90	1.000
Noz b	Nozzle	137.90	137.90	1.000
Noz d	Nozzle	137.90	137.90	1.000
Noz g1	Nozzle	137.90	137.90	1.000
Noz x-2	Nozzle	117.90	117.90	1.000
Noz g2	Nozzle	137.90	137.90	1.000
Noz x-1	Nozzle	117.90	117.90	1.000
Noz c	Nozzle	137.90	137.90	1.000
Noz h	Nozzle	137.90	137.90	1.000
Noz a2	Nozzle	117.90	117.90	1.000
Noz f	Nozzle	137.90	137.90	1.000
Noz i2	Nozzle	137.90	137.90	1.000
Minimum				1.000

Stress ratios for Pressurized Vessel Elements (MPa):

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Description	Ambient	Operating	Ratio
L' head	145.50	145.50	1.000
shell	145.50	145.50	1.000
baffle plate	142.70	142.70	1.000
shell	142.70	142.70	1.000
R'- head	145.50	145.50	1.000
Minimum			1.000

Hoop Stress in Nozzle Wall during Pressure Test (MPa):

Description	Ambient	Operating	Ratio
Noz i1	12.24	223.40	0.055
Noz a1	4.14	217.19	0.019
Noz b	7.12	223.40	0.032
Noz d	10.37	223.40	0.046
Noz g1	6.54	223.40	0.029
Noz x-2	8.14	217.20	0.037
Noz g2	6.54	223.40	0.029
Noz x-1	8.14	217.20	0.037
Noz c	9.49	223.40	0.042
Noz h	8.61	223.40	0.039
Noz a2	4.14	217.19	0.019
Noz f	10.38	223.40	0.046
Noz i2	12.24	223.40	0.055

Elements Suitable for Test Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg	Element ID Volume ltr	Corroded Metal Wgt. kg	Corroded ID Volume ltr	Extra due Misc % kg
10	20	13.2386	15.9072	11.0322	16.1024	...
20	30	47.0327	112.305	39.2798	113.305	...
30	40	38.9785	...	37.6792	...	...
40	50	5.32949	12.7257	4.45098	12.8391	...
50	60	13.2386	15.9072	11.0322	16.1024	...
Total		117	156.84	103	158.35	0

Weight of Details:

From	Type	Weight of Detail kg	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset Dtl. Cent. mm	Description
10	Nozl	0.081708	-112.5	...	...	Noz i1
20	Sadl	6.85444	40	272	...	L- Sadl
20	Sadl	6.85444	706	272	...	Sdl 2 Fr20
20	Nozl	0.044485	406.5	-232.798	...	Noz a1
20	Nozl	0.062083	541.5	214.993	-103.248	Noz b
20	Nozl	0.076443	241.5	214.559	-116.191	Noz d
20	Nozl	5.31284	201.5	208.844	113.095	Noz g1
20	Wght	100	80	231	...	WGHT:[1 OF 1]
20	Wght	100	406.5	231	...	WGHT:[2 OF 2]
40	Nozl	0.15866	35	191.879	146.2	Noz x-2
40	Nozl	2.15842	35	237.5	...	Noz g2
40	Nozl	0.15867	35	226.303	-83.5414	Noz x-1
40	Nozl	0.069796	35	...	242.5	Noz c
40	Nozl	2.27978	35	151.048	-187.791	Noz h
40	Nozl	0.050311	35	-232.798	...	Noz a2
40	Nozl	0.076413	35	-110.909	217.336	Noz f
50	Nozl	0.081708	137.5	...	...	Noz i2

Total Weight of Each Detail Type:

Saddles	13.7
Nozzles	10.6
Weights	200.0
Sum of the Detail Weights	
	224.3 kg

Weight Summation Results: (kg)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	117.8	117.8	117.8	117.8	117.8	117.8
Saddles	13.7	13.7	13.7	13.7	13.7	13.7
Nozzles	10.6	10.6	10.6	10.6	10.6	10.6
Wld Weights	200.0	200.0	200.0	200.0	200.0	200.0
Test Liquid	...	156.7	...	...	...	...
Totals	342.1	498.9	342.1	342.1	342.1	342.1

Weight Summary:

Fabricated Wt.	- Bare Weight without Removable Internals	342.1 kg
Shop Test Wt.	- Fabricated Weight + Water (Full)	498.9 kg
Shipping Wt.	- Fab. Weight + removable Intls.+ Shipping App.	342.1 kg

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Erected Wt.	- Fab. Wt + or - loose items (trays,platforms etc.)	342.1 kg
Ope. Wt. no Liq	- Fab. Weight + Internals. + Details + Weights	342.1 kg
Operating Wt.	- Empty Weight + Operating Liq. Uncorroded	342.1 kg
Oper. Wt. + CA	- Corr Wt. + Operating Liquid	327.8 kg
Field Test Wt.	- Empty Weight + Water (Full)	498.9 kg

Note:
 The Corroded Weight and thickness are used in the Horizontal
 Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Outside Surface Areas of Elements:

From	To	Surface Area mm^2
10	20	267656
20	30	1024699
30	40	...
40	50	116113
50	60	267656
Total		1676123.875 mm^2

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Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	398.000 mm
Center of Gravity of the Nozzles	498.589 mm
Center of Gravity of the Added Weights (Operating)	268.250 mm
Center of Gravity of the Added Weights (Empty)	268.250 mm
Center of Gravity of Bare Shell New and Cold	531.242 mm
Center of Gravity of Bare Shell Corroded	541.927 mm
Vessel CG in the Operating Condition	367.524 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	371.156 mm
Vessel CG in the Test Condition	387.595 mm

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Warning - Saddle Bearing Angle is less than 120 degrees - 4.15.3.1

Note:

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

Minimum Wear Plate Angle [theta1]:
 = 102.0 degrees

Note:

Wear Pad Width (.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$ and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{Rm \cdot t}, 2a)$
 = $\min(110.0 + 1.56 \cdot \sqrt{228.5 \cdot 5.0}, 2 \cdot 40.0)$
 = 80.0000 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	228.50	mm
Shell Thickness used in this Case	t	6.000	mm
Stiffened Vessel Length per 4.15.6	L	731.00	mm
Distance from Saddle to Vessel tangent	a1 or a	40.00	mm
Saddle Width	b1 or b	110.00	mm
Saddle Bearing Angle	delta or theta	90.00	degrees
Inside Depth of Head	Hi or h2	113.50	mm
Shell Allowable Stress used in Calculation		145.50	MPa
Head Allowable Stress used in Calculation		145.50	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	330.00	mm
Coefficient of Friction	mu	0.00	
Saddle Force Q, Operating Case		1687.05	N
Pressure used in Analysis	P	1.440	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	32.63	145.50
Long. Stress at Bottom of Midspan	33.18	145.50
Long. Stress at Top of Saddles	32.88	145.50
Long. Stress at Bottom of Saddles	32.93	145.50
Tangential Shear in Shell	2.41	116.40
Tangential Shear in Head	3.01	116.40
Circ. Stress at Horn of Saddle	5.75	181.88
Addl. Stress in Head as Stiffener	82.03	181.88
Circ. Compressive Stress in Shell	0.19	145.50

Load Combination Results for Q + Wind or Seismic [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 1687 + max(0, 0, 0, 0)

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$$= 1687.0 \text{ N}$$

Summary of Loads at the base of this Saddle:

Vertical Load (including saddle weight)	1754.26	N
Transverse Shear Load Saddle	0.00	N
Longitudinal Shear Load Saddle	0.00	N

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\text{theta}/2), 2*R_m)$$

$$= \min(2(450.0/2 + 6.0 + 5.0)\sin(90.0/2), 2*228.5)$$

$$= 333.754 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.0655	K2 = 1.8831	K3 = 1.6332	K4 = 0.5546
K5 = 0.9134	K6 = 0.0826	K7 = 0.0207	K8 = 0.3586
K9 = 0.2922	K10 = 0.0768	K1* = 0.1219	

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.3 [M1]:

$$= -Q*a [1 - (1 - a/L + (R_m^2 - h^2)/(2a*L))/(1 + (4h^2)/3L)]$$

$$= -1687*40.0 [1 - (1 - 40.0/731.0 + (228.5^2 - 113.5^2)/(2*40.0*731.0))/(1 + (4*113.5)/(3*731.0))]$$

$$= 22976.0 \text{ N-mm}$$

Moment per Equation 4.15.4 [M2]:

$$= Q*L/4(1 + 2(R_m^2 - h^2)/(L^2))/(1 + (4h^2)/(3L)) - 4a/L$$

$$= 1687*731/4(1 + 2(228.5^2 - 113.5^2)/(731^2))/(1 + (4*113.5)/(3*731)) - 4*40/731$$

$$= 225638.4 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.6) [Sigma1]:

$$= P * R_m/(2t) - M2/(\pi * R_m^2 * t)$$

$$= 1.44 * 228.5/(2*5.0) - 225638.4/(\pi * 228.5^2 * 5.0)$$

$$= 32.63 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.7) [Sigma2]:

$$= P * R_m/(2t) + M2/(\pi * R_m^2 * t)$$

$$= 1.44 * 228.5/(2 * 5.0) + 225638.4/(\pi * 228.5^2 * 5.0)$$

$$= 33.18 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [Sigma3]:

$$= P * R_m/(2t) - M1/(\pi * R_m^2 * t)$$

$$= 1.44 * 228.5/(2 * 5.0) - 22976.0/(\pi * 228.5^2 * 5.0)$$

$$= 32.88 \text{ MPa}$$

Longitudinal Stress at bottom of Shell at Support (4.15.9) [Sigma4]:

$$= P * R_m/(2t) + M1/(\pi * R_m^2 * t)$$

$$= 1.44 * 228.5/(2*5.0) + 22976.0/(\pi * 228.5^2 * 5.0)$$

$$= 32.93 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.5) [T]:

$$= Q(L - 2a)/(L + (4*h^2/3))$$

$$= 1687(731.0 - 2 * 40.0)/(731.0 + (4 * 113.5/3))$$

$$= 1244.7 \text{ N}$$

Shear Stress in the shell no rings, stiffened (4.15.15) [tau3]:

$$= K3 * Q / (R_m * t)$$

$$= 1.6332 * 1687/(228.5 * 5.0)$$

$$= 2.41 \text{ MPa}$$

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Shear Stress in the head, shell stiffened (4.15.16) [τ_3]:

$$= K_3 * Q / (R_m * t_h)$$

$$= 1.6332 * 1687 / (228.5 * 4.0)$$

$$= 3.01 \text{ MPa}$$

Membrane stress in the head as a stiffener (4.15.18) [σ_5]:

$$= K_4 * Q / (R_m * t_h) + ((P * R_i) / (2 * t_h)) * (R_i / h_2)$$

$$= 0.5546 * 1687 / (228.5 * 4.0) +$$

$$((1.44 * 226.0) / (2 * 4.0)) * (226.0 / 113.5)$$

$$= 82.03 \text{ MPa}$$

Decay Length (4.15.22) [x_1, x_2]:

$$= 0.78 * \sqrt{ R_m * t })$$

$$= 0.78 * \sqrt{ 228.5 * 5.0 })$$

$$= 26.365 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.23) [σ_6]:

$$= -K_5 * Q * k / (t * (b + X_1 + X_2))$$

$$= -0.9134 * 1687 * 0.1 / (5.0 * (110.0 + 26.36 + 26.36))$$

$$= -0.19 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, $L < 8R_m$ (4.15.25) [σ_7]:

$$= -Q / (4 * t * (b + X_1 + X_2)) - 12 * K_7 * Q * R_m / (L * t^2)$$

$$= -1687 / (4 * 5.0 * (110.0 + 26.365 + 26.365)) -$$

$$12 * 0.021 * 1687 * 228.5 / (731.0 * 5.0^2)$$

$$= -5.75 \text{ MPa}$$

Effective reinforcing plate width (4.15.1) [B_1]:

$$= \min(b + 1.56 * \sqrt{ R_m * t }, 2a)$$

$$= \min(110.0 + 1.56 * \sqrt{ 228.5 * 5.0 }, 2 * 40.0)$$

$$= 80.00 \text{ mm}$$

Distance between Saddle Supports [L_s]:

$$= 666.0 \text{ mm}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha * L_s (\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.000012 * 666.0 (150.0 - 21.1)$$

$$= 1.067 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	400.0000	mm
Baseplate Thickness	Bpthk	5.0000	mm
Baseplate Width	Bpwid	110.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	2	
Rib Thickness	Ribtk	5.0000	mm
Web Thickness	Webtk	5.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	239.9	MPa
Height of Web at Center	Hw,c	110.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 * \sqrt{ R * t }) + \text{WearPlateWidth}) t^3 / 12$$

$$= ((1.56 * \sqrt{ 226.0 * 5.0 }) + 90.0) 5.0^3 / 12$$

$$= 1483.750 \text{ mm}^4$$

Resolved Inertia for the Shell [I_{shell}]:

$$= \text{ShellInertia} + A_{\text{shell}} (C_1 - Y)^2$$

$$= 1483.75 + 712.201 (39.511 - 2.5)^2$$

$$= 977061.250 \text{ mm}^4$$

$$B \mid D \mid Y \mid A \mid AY \mid I + AD^2 \mid$$

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	mm	mm	mm	mm^2	mm^3	mm^4
Shell	142.4	5.0	2.5	712.2	1780.5	0.977E+06
Wearplate	90.0	5.0	7.5	450.0	3375.0	0.462E+06
Web	5.0	89.0	54.5	445.0	24252.5	0.394E+06
BasePlate	110.0	5.0	101.5	550.0	55825.0	0.211E+07
Totals	...	...	...	2157.2	85233.0	0.395E+07

Distance to Centroid [C1]:
 = AY / A
 = $3355.629 / 2157.2$
 = 39.511 mm

Angle [beta]:
 = $180 - \text{Saddle Angle} / 2$
 = $180 - 90.0 / 2$
 = 135.0

Saddle Splitting Coefficient [K1]:
 = $(1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 = $(1 + \cos(135.0) - 0.5 \sin(135.0)^2) / (\pi - 2.356 + \sin(135.0) \cos(135.0))$
 = 0.1503

Saddle Splitting Force [Fh]:
 = $K1 * Q$
 = $0.15 * 1687.049$
 = 253.5510 N

Tension Stress, $St = (Fh / As) = 0.1755$ MPa
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 143.9676$ MPa

Saddle Splitting Dimension [d]:
 = $B - R * \sin(\theta/2) / (\theta/2 \text{ in radians})$
 = $330.0 - 226.0 * \sin(90.0/2) / 0.7854$
 = 126.529 mm

Bending Moment, $M = Fh * d = 32094.4414$ N-mm

Bending Stress, $Sb = (M * C1 / I) = 0.3211$ MPa
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 159.9640$ MPa

Minimum Thickness of Baseplate per Moss:
 = $(3(Q + \text{Saddle_Wt}) \text{BasePlateWidth} / (2 * \text{BasePlateLength} * \text{AllStress}))^{.5}$
 = $(3(1687 + 67)110.0 / (2 * 400.0 * 159.964))^{.5}$
 = 2.127 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 = $2 * \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 = $2 * \cos(90 - 90.0/2) (225.0 + 6.0 + 5.0)$
 = 333.754 mm

Distance between Ribs [e]:
 = $\text{Web Length} / (\text{Nr ribs} - 1)$
 = $333.7544 / (2 - 1)$
 = 333.754 mm

Baseplate Pressure Area [Ap]:
 = $e * \text{Bpwid} / 2$
 = $333.7544 * 110.0 / 2$
 = 18356.492 mm^2

Bearing Pressure [Bp]:
 = $Q / (\text{BasePlateLength} * \text{BasePlateWidth})$

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$$= 1687.05 / (400.0 * 110.0)$$

$$= 0.038 \text{ N/mm}^2$$

Axial Load [P]:

$$= A_p * B_p$$

$$= 18356.5 * 0.04$$

$$= 703.825 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 525.0 + 834.386$$

$$= 1359.386 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 703.8 / 1359.3859$$

$$= 0.518 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	5.0	105.0	57.5	525.0	30187.5	0.108E+07
Web	166.9	5.0	2.5	834.4	2086.0	0.378E+06
Totals	...	...	...	1359.4	32273.5	0.146E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 110.0 - 5.0$$

$$= 105.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 32273.457 / 1359.386$$

$$= 23.741 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(y_{tot}, \text{Saddle Width} - y_{tot})$$

$$= \max(23.741, 110.0 - 23.741)$$

$$= 86.259 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 1458866 / 1359.386 }$$

$$= 32.759 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta / 2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 330.0 - \cos(90.0 / 2) (225.0 + 6.0 + 5.0) - 5.0$$

$$= 158.123 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ 2 * \pi^2 * \text{Elastic Modulus} / \text{Yield Stress} }$$

$$= \sqrt{ 2 * \pi^2 * 199955 / 239.9 }$$

$$= 128.255$$

Slenderness ratio [KL/r]:

$$= KL / r$$

$$= 1 * 158.123 / 32.759$$

$$= 4.827$$

Bending Moment [Rm]:

$$= F_l / (2 * B_{plen}) * e * L / 2$$

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$$= 0.0 / (2 * 400.0) * 333.754 * 158.12 / 2$$

$$= 0.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ (4.8268 < 128.255) per AISC E2-1 [Sca]:

$$= (1 - (Klr)^2 / (2 * Cc^2)) Fy / (5/3 + 3 * (Klr) / (8 * Cc) - (Klr^3) / (8 * Cc^3))$$

$$= (1 - (4.83)^2 / (2 * 128.25^2)) 240 / (5/3 + 3 * (4.83) / (8 * 128.25) - (4.83^3) / (8 * 128.25^3))$$

$$= 142.7 \text{ MPa}$$

AISC Unity Check of Outside Ribs (must be <= 1)

$$= Sc / Sca + (Rm * C1 / I) / Sba$$

$$= 0.52 / 142.66 + (0.0 * 86.259 / 1458867) / 159.96$$

$$= 0.004$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Warning - Saddle Bearing Angle is less than 120 degrees - 4.15.3.1

Note:

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

Minimum Wear Plate Angle [theta1]:

$$= 102.0 \text{ degrees}$$

Note:

Wear Pad Width (.00) is less than $1.56 * \sqrt{rm * t}$ and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:

$$= \min(b + 1.56 * \sqrt{ Rm * t }, 2a)$$

$$= \min(110.0 + 1.56 * \sqrt{ 228.5 * 5.0 }, 2 * 40.0)$$

$$= 80.0000 \text{ mm}$$

Input and Calculated Values:

Vessel Mean Radius	Rm	228.50	mm
Shell Thickness used in this Case	t	6.000	mm
Stiffened Vessel Length per 4.15.6	L	731.00	mm
Distance from Saddle to Vessel tangent	a1 or a	40.00	mm
Saddle Width	b1 or b	110.00	mm
Saddle Bearing Angle	delta or theta	90.00	degrees
Shell Allowable Stress used in Calculation		145.50	MPa
Head Allowable Stress used in Calculation		142.70	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	330.00	mm
Coefficient of Friction	mu	0.00	
Saddle Force Q, Operating Case		1392.87	N
Pressure used in Analysis	P	1.440	MPa
Horizontal Vessel Analysis Results:	Actual Allowable		
	MPa MPa		
Long. Stress at Top of Midspan	32.60	145.50	
Long. Stress at Bottom of Midspan	33.21	145.50	
Long. Stress at Top of Saddles	32.85	145.50	
Long. Stress at Bottom of Saddles	32.96	145.50	
Tangential Shear in Shell	1.99	116.40	

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Tangential Shear in Head	0.34		114.16	
Circ. Stress at Horn of Saddle	4.74		181.88	
Circ. Compressive Stress in Shell	0.16		145.50	

Load Combination Results for Q + Wind or Seismic [Q]:
 = Saddle Load + max(FwL, Fwt, Fsl, Fst)
 = 1393 + max(0, 0, 0, 0)
 = 1392.9 N

Summary of Loads at the base of this Saddle:

Vertical Load (including saddle weight)	1460.09	N
Transverse Shear Load Saddle	0.00	N
Longitudinal Shear Load Saddle	0.00	N

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness}) \sin(\text{theta}/2), 2 * R_m)$
 $= \min(2(450.0/2 + 6.0 + 5.0) \sin(90.0/2), 2 * 228.5)$
 $= 333.754 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.0655	K2 = 1.8831	K3 = 1.6332	K4 = 0.5546
K5 = 0.9134	K6 = 0.0826	K7 = 0.0207	K8 = 0.3586
K9 = 0.2922	K10 = 0.0768	K1* = 0.1219	

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.3 [M1]:
 $= -Q * a [1 - (1 - a/L + (R_m^2 - h^2)/(2a * L)) / (1 + (4h^2)/3L)]$
 $= -1393 * 40.0 [1 - (1 - 40.0/731.0 + (228.5^2 - 0.0^2)/(2 * 40.0 * 731.0)) / (1 + (4 * 0.0)/(3 * 731.0))]$
 $= 46713.8 \text{ N-mm}$

Moment per Equation 4.15.4 [M2]:
 $= Q * L / 4 (1 + 2(R_m^2 - h^2)/(L^2)) / (1 + (4h^2)/(3L)) - 4a/L$
 $= 1393 * 731 / 4 (1 + 2(228.5^2 - 0^2)/(731^2)) / (1 + (4 * 0)/(3 * 731)) - 4 * 40 / 731$
 $= 248677.1 \text{ N-mm}$

Longitudinal Stress at Top of Shell (4.15.6) [Sigma1]:
 $= P * R_m / (2t) - M2 / (\pi * R_m^2 * t)$
 $= 1.44 * 228.5 / (2 * 5.0) - 248677.1 / (\pi * 228.5^2 * 5.0)$
 $= 32.60 \text{ MPa}$

Longitudinal Stress at Bottom of Shell (4.15.7) [Sigma2]:
 $= P * R_m / (2t) + M2 / (\pi * R_m^2 * t)$
 $= 1.44 * 228.5 / (2 * 5.0) + 248677.1 / (\pi * 228.5^2 * 5.0)$
 $= 33.21 \text{ MPa}$

Longitudinal Stress at Top of Shell at Support (4.15.8) [Sigma3]:
 $= P * R_m / (2t) - M1 / (\pi * R_m^2 * t)$
 $= 1.44 * 228.5 / (2 * 5.0) - 46713.8 / (\pi * 228.5^2 * 5.0)$
 $= 32.85 \text{ MPa}$

Longitudinal Stress at bottom of Shell at Support (4.15.9) [Sigma4]:
 $= P * R_m / (2t) + M1 / (\pi * R_m^2 * t)$
 $= 1.44 * 228.5 / (2 * 5.0) + 46713.8 / (\pi * 228.5^2 * 5.0)$
 $= 32.96 \text{ MPa}$

Maximum Shear Force in the Saddle (4.15.5) [T]:
 $= Q(L - 2a) / (L + (4 * h^2/3))$

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$$= 1393(731.0 - 2 * 40.0)/(731.0 + (4 * 0.0/3))$$

$$= 1240.4 \text{ N}$$

Shear Stress in the shell no rings, stiffened (4.15.15) [tau3]:

$$= K3 * Q / (Rm * t)$$

$$= 1.6332 * 1393 / (228.5 * 5.0)$$

$$= 1.99 \text{ MPa}$$

Shear Stress in the head, shell stiffened (4.15.16) [tau3*]:

$$= K3 * Q / (Rm * th)$$

$$= 1.6332 * 1393 / (228.5 * 29.0)$$

$$= 0.34 \text{ MPa}$$

Decay Length (4.15.22) [x1,x2]:

$$= 0.78 * \sqrt{Rm * t}$$

$$= 0.78 * \sqrt{228.5 * 5.0}$$

$$= 26.365 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.23) [sigma6]:

$$= -K5 * Q * k / (t * (b + X1 + X2))$$

$$= -0.9134 * 1393 * 0.1 / (5.0 * (110.0 + 26.36 + 26.36))$$

$$= -0.16 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.25) [sigma7*]:

$$= -Q/(4*t*(b+X1+X2)) - 12*K7*Q*Rm/(L*t^2)$$

$$= -1393/(4*5.0*(110.0+26.365+26.365)) -$$

$$12 * 0.021 * 1393 * 228.5 / (731.0 * 5.0^2)$$

$$= -4.74 \text{ MPa}$$

Effective reinforcing plate width (4.15.1) [B1]:

$$= \min(b + 1.56 * \sqrt{Rm * t}, 2a)$$

$$= \min(110.0 + 1.56 * \sqrt{228.5 * 5.0}, 2 * 40.0)$$

$$= 80.00 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	400.0000	mm
Baseplate Thickness	Bpthk	5.0000	mm
Baseplate Width	Bpwid	110.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	2	
Rib Thickness	Ribtk	5.0000	mm
Web Thickness	Webtk	5.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	239.9	MPa
Height of Web at Center	Hw,c	110.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 * \sqrt{R * t}) + \text{WearPlateWidth})^3 / 12$$

$$= ((1.56 * \sqrt{226.0 * 5.0}) + 90.0)^3 / 12$$

$$= 1483.750 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 1483.75 + 712.201(39.511 - 2.5)^2$$

$$= 977061.250 \text{ mm}^4$$

	B mm	D mm	Y mm	A mm^2	AY mm^3	I + AD^2 mm^4
Shell	142.4	5.0	2.5	712.2	1780.5	0.977E+06
Wearplate	90.0	5.0	7.5	450.0	3375.0	0.462E+06
Web	5.0	89.0	54.5	445.0	24252.5	0.394E+06

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BasePlate	110.0	5.0	101.5	550.0	55825.0	0.211E+07
Totals	...	...	...	2157.2	85233.0	0.395E+07

Distance to Centroid [C1]:

= AY / A
 = $3355.629/2157.2$
 = 39.511 mm

Angle [beta]:

= $180 - \text{Saddle Angle}/2$
 = $180 - 90.0/2$
 = 135.0

Saddle Splitting Coefficient [K1]:

= $(1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 = $(1 + \cos(135.0) - 0.5 \sin(135.0)^2) / (\pi - 2.356 + \sin(135.0) \cos(135.0))$
 = 0.1503

Saddle Splitting Force [Fh]:

= $K1 * Q$
 = $0.15 * 1392.874$
 = 209.3386 N

Tension Stress, $St = (Fh/As) = 0.1449$ MPa
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 143.9676$ MPa

Saddle Splitting Dimension [d]:

= $B - R * \sin(\theta/2) / (\theta/2 \text{ in radians})$
 = $330.0 - 226.0 * \sin(90.0/2) / 0.7854$
 = 126.529 mm

Bending Moment, $M = Fh * d = 26498.0449$ N-mm

Bending Stress, $Sb = (M * C1 / I) = 0.2651$ MPa
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 159.9640$ MPa

Minimum Thickness of Baseplate per Moss:

= $(3(Q + \text{Saddle_Wt}) \text{BasePlateWidth} / (2 * \text{BasePlateLength} * \text{AllStress}))^{.5}$
 = $(3(1393 + 67)110.0 / (2 * 400.0 * 159.964))^{.5}$
 = 1.940 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

= $2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 = $2 * \cos(90 - 90.0/2) (225.0 + 6.0 + 5.0)$
 = 333.754 mm

Distance between Ribs [e]:

= $\text{Web Length} / (N_{\text{ribs}} - 1)$
 = $333.7544 / (2 - 1)$
 = 333.754 mm

Baseplate Pressure Area [Ap]:

= $e * \text{Bpwid} / 2$
 = $333.7544 * 110.0/2$
 = 18356.492 mm²

Bearing Pressure [Bp]:

= $Q / (\text{BasePlateLength} * \text{BasePlateWidth})$
 = $1392.874 / (400.0 * 110.0)$
 = 0.032 N/mm²

Axial Load [P]:

= $Ap * Bp$

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$$= 18356.5 * 0.03$$

$$= 581.097 \text{ N}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 525.0 + 834.386$$

$$= 1359.386 \text{ mm}^2$$

Compressive Stress [Sc]:

$$= P/Ar$$

$$= 581.1/1359.3859$$

$$= 0.428 \text{ MPa}$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	5.0	105.0	57.5	525.0	30187.5	0.108E+07
Web	166.9	5.0	2.5	834.4	2086.0	0.378E+06
Totals	...	...	...	1359.4	32273.5	0.146E+07

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 110.0 - 5.0$$

$$= 105.000 \text{ mm}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 32273.457/1359.386$$

$$= 23.741 \text{ mm}$$

Distance to Centroid [C1]:

$$= \max(\text{ytot}, \text{Saddle Width} - \text{ytot})$$

$$= \max(23.741, 110.0 - 23.741)$$

$$= 86.259 \text{ mm}$$

Radius of Gyration [r]:

$$= \sqrt{ \text{Total Inertia} / \text{Total Area} }$$

$$= \sqrt{ 1458866/1359.386 }$$

$$= 32.759 \text{ mm}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 330.0 - \cos(90.0/2) (225.0 + 6.0 + 5.0) - 5.0$$

$$= 158.123 \text{ mm}$$

Intermediate Term [Cc]:

$$= \sqrt{ 2 * \pi^2 * \text{Elastic Modulus} / \text{Yield Stress} }$$

$$= \sqrt{ 2 * \pi^2 * 199955/239.9 }$$

$$= 128.255$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 * 158.123/32.759$$

$$= 4.827$$

Bending Moment [Rm]:

$$= FL / (2 * Bplen) * e * L / 2$$

$$= 0.0 / (2 * 400.0) * 333.754 * 158.12/2$$

$$= 0.000 \text{ N-mm}$$

Compressive Allowable, $KL/r < Cc$ ($4.8268 < 128.255$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 * Cc^2)) Fy / (5/3 + 3 * (KL/r) / (8 * Cc) - (KL/r)^3 / (8 * Cc^3))$$

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$$\begin{aligned} &= (1 - (4.83)^2 / (2 * 128.25^2)) 240 / \\ &\quad (5/3 + 3 * (4.83) / (8 * 128.25) - (4.83^3) / (8 * 128.25^3)) \\ &= 142.7 \text{ MPa} \end{aligned}$$

AISC Unity Check of Outside Ribs (must be <= 1)

$$\begin{aligned} &= Sc/Sca + (Rm * C1 / I) / Sba \\ &= 0.43/142.66 + (0.0 * 86.259/1458867) / 159.96 \\ &= 0.003 \end{aligned}$$

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Warning - Saddle Bearing Angle is less than 120 degrees - 4.15.3.1

Note:

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

Minimum Wear Plate Angle [theta1]:
 = 102.0 degrees

Note:

Wear Pad Width (.00) is less than $1.56 \cdot \sqrt{r_m \cdot t}$ and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{R_m \cdot t}, 2a)$
 = $\min(110.0 + 1.56 \cdot \sqrt{228.0 \cdot 6.0}, 2 \cdot 40.0)$
 = 80.0000 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	228.00	mm
Shell Thickness used in this Case	t	6.000	mm
Stiffened Vessel Length per 4.15.6	L	731.00	mm
Distance from Saddle to Vessel tangent	a1 or a	40.00	mm
Saddle Width	b1 or b	110.00	mm
Saddle Bearing Angle	delta or theta	90.00	degrees
Inside Depth of Head	Hi or h2	112.50	mm
Shell Allowable Stress used in Calculation		327.51	MPa
Head Allowable Stress used in Calculation		327.51	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	330.00	mm
Coefficient of Friction	mu	0.00	
Saddle Force Q, Test Case, no Ext. Forces		2455.26	N
Pressure used in Analysis	P	2.061	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	38.83	327.51
Long. Stress at Bottom of Midspan	39.50	327.51
Long. Stress at Top of Saddles	39.13	327.51
Long. Stress at Bottom of Saddles	39.20	327.51
Tangential Shear in Shell	2.93	183.75
Tangential Shear in Head	3.52	162.72
Circ. Stress at Horn of Saddle	5.88	491.27
Addl. Stress in Head as Stiffener	93.96	327.51
Circ. Compressive Stress in Shell	0.22	327.51

Load Combination Results for Q + Wind or Seismic [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 2455 + max(0, 0, 0, 0)

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$$= 2455.3 \text{ N}$$

Summary of Loads at the base of this Saddle:

Vertical Load (including saddle weight)	2522.47	N
Transverse Shear Load Saddle	0.00	N
Longitudinal Shear Load Saddle	0.00	N

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\text{theta}/2), 2*R_m)$$

$$= \min(2(450.0/2 + 6.0 + 5.0)\sin(90.0/2), 2*228.0)$$

$$= 333.754 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.0655	K2 = 1.8831	K3 = 1.6332	K4 = 0.5546
K5 = 0.9134	K6 = 0.0826	K7 = 0.0207	K8 = 0.3586
K9 = 0.2922	K10 = 0.0768	K1* = 0.1219	

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.3 [M1]:

$$= -Q*a[1 - (1 - a/L + (R_m^2 - h^2)/(2a*L))/(1 + (4h^2)/3L)]$$

$$= -2455*40.0[1 - (1 - 40.0/731.0 + (228.0^2 - 112.5^2)/(2*40.0*731.0))/(1 + (4*112.5)/(3*731.0))]$$

$$= 33634.4 \text{ N-mm}$$

Moment per Equation 4.15.4 [M2]:

$$= Q*L/4(1 + 2(R_m^2 - h^2)/(L^2))/(1 + (4h^2)/(3L)) - 4a/L$$

$$= 2455*731/4(1 + 2(228^2 - 113^2)/(731^2))/(1 + (4*113)/(3*731)) - 4*40/731$$

$$= 329026.4 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.6) [Sigma1]:

$$= P * R_m/(2t) - M2/(\pi * R_m^2 * t)$$

$$= 2.061 * 228.0/(2*6.0) - 329026.4/(\pi * 228.0^2 * 6.0)$$

$$= 38.83 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.7) [Sigma2]:

$$= P * R_m/(2t) + M2/(\pi * R_m^2 * t)$$

$$= 2.061 * 228.0/(2 * 6.0) + 329026.4/(\pi * 228.0^2 * 6.0)$$

$$= 39.50 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [Sigma3]:

$$= P * R_m/(2t) - M1/(\pi * R_m^2 * t)$$

$$= 2.061 * 228.0/(2 * 6.0) - 33634.4/(\pi * 228.0^2 * 6.0)$$

$$= 39.13 \text{ MPa}$$

Longitudinal Stress at bottom of Shell at Support (4.15.9) [Sigma4]:

$$= P * R_m/(2t) + M1/(\pi * R_m^2 * t)$$

$$= 2.061 * 228.0/(2*6.0) + 33634.4/(\pi * 228.0^2 * 6.0)$$

$$= 39.20 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.5) [T]:

$$= Q(L - 2a)/(L + (4h^2/3))$$

$$= 2455(731.0 - 2 * 40.0)/(731.0 + (4 * 112.5/3))$$

$$= 1814.3 \text{ N}$$

Shear Stress in the shell no rings, stiffened (4.15.15) [tau3]:

$$= K3 * Q / (R_m * t)$$

$$= 1.6332 * 2455/(228.0 * 6.0)$$

$$= 2.93 \text{ MPa}$$

Shear Stress in the head, shell stiffened (4.15.16) [tau3*]:

$$= K3 * Q / (Rm * th)$$

$$= 1.6332 * 2455 / (228.0 * 5.0)$$

$$= 3.52 \text{ MPa}$$

Membrane stress in the head as a stiffener (4.15.18) [sigma5]:

$$= K4 * Q / (Rm * th) + ((P * Ri) / (2 * th)) * (Ri / h2)$$

$$= 0.5546 * 2455 / (228.0 * 5.0) +$$

$$((2.06 * 225.0) / (2 * 5.0)) * (225.0 / 112.5)$$

$$= 93.96 \text{ MPa}$$

Decay Length (4.15.22) [x1,x2]:

$$= 0.78 * \text{sqrt}(Rm * t)$$

$$= 0.78 * \text{sqrt}(228.0 * 6.0)$$

$$= 28.849 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.23) [sigma6]:

$$= -K5 * Q * k / (t * (b + X1 + X2))$$

$$= - 0.9134 * 2455 * 0.1 / (6.0 * (110.0 + 28.85 + 28.85))$$

$$= -0.22 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.25) [sigma7*]:

$$= -Q / (4 * t * (b + X1 + X2)) - 12 * K7 * Q * Rm / (L * t^2)$$

$$= -2455 / (4 * 6.0 * (110.0 + 28.849 + 28.849)) -$$

$$12 * 0.021 * 2455 * 228.0 / (731.0 * 6.0^2)$$

$$= -5.88 \text{ MPa}$$

Effective reinforcing plate width (4.15.1) [B1]:

$$= \min(b + 1.56 * \text{sqrt}(Rm * t), 2a)$$

$$= \min(110.0 + 1.56 * \text{sqrt}(228.0 * 6.0), 2 * 40.0)$$

$$= 80.00 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	400.0000	mm
Baseplate Thickness	Bpthk	5.0000	mm
Baseplate Width	Bpwid	110.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	2	
Rib Thickness	Ribtk	5.0000	mm
Web Thickness	Webtk	5.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	239.9	MPa
Height of Web at Center	Hw,c	110.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 * \text{sqrt}(R * t) + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 * \text{sqrt}(225.0 * 6.0) + 90.0) 6.0^3) / 12$$

$$= 2651.723 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 2651.723 + 883.908 (37.592 - 3.0)^2$$

$$= 1060364.875 \text{ mm}^4$$

	B	D	Y	A	AY	I + AD ²
	mm	mm	mm	mm ²	mm ³	mm ⁴
Shell	147.3	6.0	3.0	883.9	2651.7	0.106E+07
Wearplate	90.0	5.0	8.5	450.0	3825.0	0.382E+06
Web	5.0	89.0	55.5	445.0	24697.5	0.436E+06
BasePlate	110.0	5.0	102.5	550.0	56375.0	0.232E+07
Totals	...	...	...	2328.9	87549.2	0.420E+07

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Distance to Centroid [C1]:
 = AY / A
 = $3446.82 / 2328.908$
 = 37.592 mm

Angle [beta]:
 = $180 - \text{Saddle Angle} / 2$
 = $180 - 90.0 / 2$
 = 135.0

Saddle Splitting Coefficient [K1]:
 = $(1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$
 = $(1 + \cos(135.0) - 0.5 \sin(135.0)^2) / (\pi - 2.356 + \sin(135.0) \cos(135.0))$
 = 0.1503

Saddle Splitting Force [Fh]:
 = $K1 * Q$
 = $0.15 * 2455.256$
 = 369.0067 N

Tension Stress, $St = (Fh / As) = 0.2554$ MPa
 Allowed Stress, $Sa = 0.6 * \text{Yield Str} = 143.9676$ MPa

Saddle Splitting Dimension [d]:
 = $B - R * \sin(\theta/2) / (\theta/2 \text{ in radians})$
 = $330.0 - 225.0 * \sin(90.0/2) / 0.7854$
 = 127.429 mm

Bending Moment, $M = Fh * d = 47041.1680$ N-mm

Bending Stress, $Sb = (M * C1 / I) = 0.4212$ MPa
 Allowed Stress, $Sa = 2/3 * \text{Yield Str} = 159.9640$ MPa

Minimum Thickness of Baseplate per Moss:

= $(3(Q + \text{Saddle_Wt}) \text{BasePlateWidth} / (2 * \text{BasePlateLength} * \text{AllStress}))^{.5}$
 = $(3(2455 + 67)110.0 / (2 * 400.0 * 159.964))^{.5}$
 = 2.551 mm

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 = $2 * \cos(90 - \text{Saddle Angle} / 2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 = $2 * \cos(90 - 90.0/2) (225.0 + 6.0 + 5.0)$
 = 333.754 mm

Distance between Ribs [e]:
 = $\text{Web Length} / (N_{\text{ribs}} - 1)$
 = $333.7544 / (2 - 1)$
 = 333.754 mm

Baseplate Pressure Area [Ap]:
 = $e * B_{\text{pwid}} / 2$
 = $333.7544 * 110.0 / 2$
 = 18356.492 mm²

Bearing Pressure [Bp]:
 = $Q / (\text{BasePlateLength} * \text{BasePlateWidth})$
 = $2455.256 / (400.0 * 110.0)$
 = 0.056 N/mm²

Axial Load [P]:
 = $Ap * Bp$
 = $18356.5 * 0.06$
 = 1024.316 N

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Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = 525.0 + 834.386
 = 1359.386 mm²

Compressive Stress [Sc]:
 = P/Ar
 = 1024.3/1359.3859
 = 0.754 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	5.0	105.0	57.5	525.0	30187.5	0.108E+07
Web	166.9	5.0	2.5	834.4	2086.0	0.378E+06
Totals	...	...	...	1359.4	32273.5	0.146E+07

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 110.0 - 5.0
 = 105.000 mm

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 32273.457/1359.386
 = 23.741 mm

Distance to Centroid [C1]:
 = max(ytot, Saddle Width - ytot)
 = max(23.741, 110.0 - 23.741)
 = 86.259 mm

Radius of Gyration [r]:
 = sqrt(Total Inertia / Total Area)
 = sqrt(1458866/1359.386)
 = 32.759 mm

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2) (radius + shlthk + wpdthk) - bpthk
 = 330.0 - cos(90.0/2) (225.0 + 6.0 + 5.0) - 5.0
 = 158.123 mm

Intermediate Term [Cc]:
 = sqrt(2 * pi² * Elastic Modulus / Yield Stress)
 = sqrt(2 * pi² * 199955/239.9)
 = 128.255

Slenderness ratio [KL/r]:
 = KL/r
 = 1 * 158.123/32.759
 = 4.827

Bending Moment [Rm]:
 = FL / (2 * Bplen) * e * L / 2
 = 0.0 / (2 * 400.0) * 333.754 * 158.12/2
 = 0.000 N-mm

Compressive Allowable, KL/r < Cc (4.8268 < 128.255) per AISC E2-1 [Sca]:
 = (1 - (KL/r)² / (2 * Cc²)) Fy / (5/3 + 3 * (KL/r) / (8 * Cc) - (KL/r)³ / (8 * Cc³))
 = (1 - (4.83)² / (2 * 128.25²)) 240 /
 (5/3 + 3 * (4.83) / (8 * 128.25) - (4.83³) / (8 * 128.25³))
 = 142.7 MPa

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AISC Unity Check of Outside Ribs (must be <= 1)

$$= Sc/Sca + (Rm * C1 / I)/Sba$$

$$= 0.75/142.66 + (0.0 * 86.259/1458867)/159.96$$

$$= 0.005$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Warning - Saddle Bearing Angle is less than 120 degrees - 4.15.3.1

Note:

Wear Pad Bearing angle is less than saddle angle + saddle/12 degrees. The wear plate will be ignored.

Minimum Wear Plate Angle [theta1]:
 = 102.0 degrees

Note:

Wear Pad Width (.00) is less than $1.56 \cdot \sqrt{rm \cdot t}$ and less than 2a. The wear plate will be ignored.

Minimum Wear Plate Width to be considered in analysis [b1]:
 = $\min(b + 1.56 \cdot \sqrt{ Rm * t }, 2a)$
 = $\min(110.0 + 1.56 \cdot \sqrt{ 228.0 * 6.0 }, 2 * 40.0)$
 = 80.0000 mm

Input and Calculated Values:

Vessel Mean Radius	Rm	228.00	mm
Shell Thickness used in this Case	t	6.000	mm
Stiffened Vessel Length per 4.15.6	L	731.00	mm
Distance from Saddle to Vessel tangent	a1 or a	40.00	mm
Saddle Width	b1 or b	110.00	mm
Saddle Bearing Angle	delta or theta	90.00	degrees
Shell Allowable Stress used in Calculation		327.51	MPa
Head Allowable Stress used in Calculation		327.51	MPa
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	330.00	mm
Coefficient of Friction	mu	0.00	
Saddle Force Q, Test Case, no Ext. Forces		2302.40	N
Pressure used in Analysis	P	2.061	MPa

Horizontal Vessel Analysis Results:	Actual MPa	Allowable MPa
Long. Stress at Top of Midspan	38.75	327.51
Long. Stress at Bottom of Midspan	39.59	327.51
Long. Stress at Top of Saddles	39.09	327.51
Long. Stress at Bottom of Saddles	39.25	327.51
Tangential Shear in Shell	2.75	183.75
Tangential Shear in Head	0.55	153.50
Circ. Stress at Horn of Saddle	5.52	491.27
Circ. Compressive Stress in Shell	0.21	327.51

Load Combination Results for Q + Wind or Seismic [Q]:
 = Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 2302 + max(0, 0, 0, 0)

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$$= 2302.4 \text{ N}$$

Summary of Loads at the base of this Saddle:

Vertical Load (including saddle weight)	2369.61	N
Transverse Shear Load Saddle	0.00	N
Longitudinal Shear Load Saddle	0.00	N

Formulas and Substitutions for Horizontal Vessel Analysis:

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\text{theta}/2), 2R_m)$$

$$= \min(2(450.0/2 + 6.0 + 5.0)\sin(90.0/2), 2 \times 228.0)$$

$$= 333.754 \text{ mm}$$

The Computed K values from Table 4.15.1:

K1 = 0.0655	K2 = 1.8831	K3 = 1.6332	K4 = 0.5546
K5 = 0.9134	K6 = 0.0826	K7 = 0.0207	K8 = 0.3586
K9 = 0.2922	K10 = 0.0768	K1* = 0.1219	

Note: Dimension a is less than $R_m/2$.

Moment per Equation 4.15.3 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h^2)/(2a \cdot L)) / (1 + (4h^2)/3L)]$$

$$= -2302 \cdot 40.0 [1 - (1 - 40.0/731.0 + (228.0^2 - 0.0^2) / (2 \cdot 40.0 \cdot 731.0)) / (1 + (4 \cdot 0.0) / (3 \cdot 731.0))]$$

$$= 76857.6 \text{ N-mm}$$

Moment per Equation 4.15.4 [M2]:

$$= Q \cdot L / 4 (1 + 2(R_m^2 - h^2)/(L^2)) / (1 + (4h^2)/(3L)) - 4a/L$$

$$= 2302 \cdot 731 / 4 (1 + 2(228^2 - 0^2)/(731^2)) / (1 + (4 \cdot 0) / (3 \cdot 731)) - 4 \cdot 40 / 731$$

$$= 410699.8 \text{ N-mm}$$

Longitudinal Stress at Top of Shell (4.15.6) [Sigma1]:

$$= P \cdot R_m / (2t) - M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 2.061 \cdot 228.0 / (2 \cdot 6.0) - 410699.8 / (\pi \cdot 228.0^2 \cdot 6.0)$$

$$= 38.75 \text{ MPa}$$

Longitudinal Stress at Bottom of Shell (4.15.7) [Sigma2]:

$$= P \cdot R_m / (2t) + M2 / (\pi \cdot R_m^2 \cdot t)$$

$$= 2.061 \cdot 228.0 / (2 \cdot 6.0) + 410699.8 / (\pi \cdot 228.0^2 \cdot 6.0)$$

$$= 39.59 \text{ MPa}$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [Sigma3]:

$$= P \cdot R_m / (2t) - M1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 2.061 \cdot 228.0 / (2 \cdot 6.0) - 76857.6 / (\pi \cdot 228.0^2 \cdot 6.0)$$

$$= 39.09 \text{ MPa}$$

Longitudinal Stress at bottom of Shell at Support (4.15.9) [Sigma4]:

$$= P \cdot R_m / (2t) + M1 / (\pi \cdot R_m^2 \cdot t)$$

$$= 2.061 \cdot 228.0 / (2 \cdot 6.0) + 76857.6 / (\pi \cdot 228.0^2 \cdot 6.0)$$

$$= 39.25 \text{ MPa}$$

Maximum Shear Force in the Saddle (4.15.5) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h^2/3))$$

$$= 2302(731.0 - 2 \cdot 40.0) / (731.0 + (4 \cdot 0.0/3))$$

$$= 2050.4 \text{ N}$$

Shear Stress in the shell no rings, stiffened (4.15.15) [tau3]:

$$= K3 \cdot Q / (R_m \cdot t)$$

$$= 1.6332 \cdot 2302 / (228.0 \cdot 6.0)$$

$$= 2.75 \text{ MPa}$$

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Shear Stress in the head, shell stiffened (4.15.16) [tau3*]:

$$= K3 * Q / (Rm * th)$$

$$= 1.6332 * 2302 / (228.0 * 30.0)$$

$$= 0.55 \text{ MPa}$$

Decay Length (4.15.22) [x1,x2]:

$$= 0.78 * \text{sqrt}(Rm * t)$$

$$= 0.78 * \text{sqrt}(228.0 * 6.0)$$

$$= 28.849 \text{ mm}$$

Circumferential Stress in shell, no rings (4.15.23) [sigma6]:

$$= -K5 * Q * k / (t * (b + X1 + X2))$$

$$= - 0.9134 * 2302 * 0.1 / (6.0 * (110.0 + 28.85 + 28.85))$$

$$= -0.21 \text{ MPa}$$

Circ. Comp. Stress at Horn of Saddle, L<8Rm (4.15.25) [sigma7*]:

$$= -Q / (4 * t * (b + X1 + X2)) - 12 * K7 * Q * Rm / (L * t^2)$$

$$= -2302 / (4 * 6.0 * (110.0 + 28.849 + 28.849)) -$$

$$12 * 0.021 * 2302 * 228.0 / (731.0 * 6.0^2)$$

$$= -5.52 \text{ MPa}$$

Effective reinforcing plate width (4.15.1) [B1]:

$$= \min(b + 1.56 * \text{sqrt}(Rm * t), 2a)$$

$$= \min(110.0 + 1.56 * \text{sqrt}(228.0 * 6.0), 2 * 40.0)$$

$$= 80.00 \text{ mm}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	400.0000	mm
Baseplate Thickness	Bpthk	5.0000	mm
Baseplate Width	Bpwid	110.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	2	
Rib Thickness	Ribtk	5.0000	mm
Web Thickness	Webtk	5.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	239.9	MPa
Height of Web at Center	Hw,c	110.0	mm
Friction Coefficient	mu	0.000	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 * \text{sqrt}(R * t) + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 * \text{sqrt}(225.0 * 6.0) + 90.0) 6.0^3) / 12$$

$$= 2651.723 \text{ mm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{shell}} (C1 - Y)^2$$

$$= 2651.723 + 883.908 (37.592 - 3.0)^2$$

$$= 1060364.875 \text{ mm}^4$$

	B	D	Y	A	AY	I + AD ²
	mm	mm	mm	mm ²	mm ³	mm ⁴
Shell	147.3	6.0	3.0	883.9	2651.7	0.106E+07
Wearplate	90.0	5.0	8.5	450.0	3825.0	0.382E+06
Web	5.0	89.0	55.5	445.0	24697.5	0.436E+06
BasePlate	110.0	5.0	102.5	550.0	56375.0	0.232E+07
Totals	...	...	...	2328.9	87549.2	0.420E+07

Distance to Centroid [C1]:

$$= AY / A$$

$$= 3446.82 / 2328.908$$

$$= 37.592 \text{ mm}$$

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Angle [beta]:

$$\begin{aligned}
 &= 180 - \text{Saddle Angle}/2 \\
 &= 180 - 90.0/2 \\
 &= 135.0
 \end{aligned}$$

Saddle Splitting Coefficient [K1]:

$$\begin{aligned}
 &= (1 + \cos(\beta) - 0.5 \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta)) \\
 &= (1 + \cos(135.0) - 0.5 \sin(135.0)^2) / (\pi - 2.356 + \sin(135.0) \cos(135.0)) \\
 &= 0.1503
 \end{aligned}$$

Saddle Splitting Force [Fh]:

$$\begin{aligned}
 &= K1 * Q \\
 &= 0.15 * 2302.399 \\
 &= 346.0336 \text{ N}
 \end{aligned}$$

$$\begin{aligned}
 \text{Tension Stress, } St &= (Fh / As) = 0.2395 \text{ MPa} \\
 \text{Allowed Stress, } Sa &= 0.6 * \text{Yield Str} = 143.9676 \text{ MPa}
 \end{aligned}$$

Saddle Splitting Dimension [d]:

$$\begin{aligned}
 &= B - R * \sin(\theta/2) / (\theta/2 \text{ in radians}) \\
 &= 330.0 - 225.0 * \sin(90.0/2) / 0.7854 \\
 &= 127.429 \text{ mm}
 \end{aligned}$$

$$\text{Bending Moment, } M = Fh * d = 44112.5391 \text{ N-mm}$$

$$\begin{aligned}
 \text{Bending Stress, } Sb &= (M * C1 / I) = 0.3950 \text{ MPa} \\
 \text{Allowed Stress, } Sa &= 2/3 * \text{Yield Str} = 159.9640 \text{ MPa}
 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned}
 &= (3(Q + \text{Saddle_Wt}) \text{BasePlateWidth} / (2 * \text{BasePlateLength} * \text{AllStress}))^{.5} \\
 &= (3(2302 + 67)110.0 / (2 * 400.0 * 159.964))^{.5} \\
 &= 2.472 \text{ mm}
 \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$\begin{aligned}
 &= 2 * \cos(90 - \text{Saddle Angle}/2) * (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk}) \\
 &= 2 * \cos(90 - 90.0/2) * (225.0 + 6.0 + 5.0) \\
 &= 333.754 \text{ mm}
 \end{aligned}$$

Distance between Ribs [e]:

$$\begin{aligned}
 &= \text{Web Length} / (\text{Nr ribs} - 1) \\
 &= 333.7544 / (2 - 1) \\
 &= 333.754 \text{ mm}
 \end{aligned}$$

Baseplate Pressure Area [Ap]:

$$\begin{aligned}
 &= e * \text{Bpwid} / 2 \\
 &= 333.7544 * 110.0 / 2 \\
 &= 18356.492 \text{ mm}^2
 \end{aligned}$$

Bearing Pressure [Bp]:

$$\begin{aligned}
 &= Q / (\text{BasePlateLength} * \text{BasePlateWidth}) \\
 &= 2302.399 / (400.0 * 110.0) \\
 &= 0.052 \text{ N/mm}^2
 \end{aligned}$$

Axial Load [P]:

$$\begin{aligned}
 &= Ap * Bp \\
 &= 18356.5 * 0.05 \\
 &= 960.545 \text{ N}
 \end{aligned}$$

Area of the Rib and Web [Ar]:

$$\begin{aligned}
 &= \text{Rib Area} + \text{Web Area} \\
 &= 525.0 + 834.386 \\
 &= 1359.386 \text{ mm}^2
 \end{aligned}$$

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Compressive Stress [Sc]:

= P/Ar
 = 960.5/1359.3859
 = 0.707 MPa

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib	5.0	105.0	57.5	525.0	30187.5	0.108E+07
Web	166.9	5.0	2.5	834.4	2086.0	0.378E+06
Totals	...	...	...	1359.4	32273.5	0.146E+07

Rib dimension [D]:

= Saddle Width - Web Thickness
 = 110.0 - 5.0
 = 105.000 mm

Distance to Centroid from Datum [ytot]:

= AY / A
 = 32273.457/1359.386
 = 23.741 mm

Distance to Centroid [C1]:

= max(ytot, Saddle Width - ytot)
 = max(23.741, 110.0 - 23.741)
 = 86.259 mm

Radius of Gyration [r]:

= sqrt(Total Inertia / Total Area)
 = sqrt(1458866/1359.386)
 = 32.759 mm

Length of Outer Rib [L]:

= Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 330.0 - cos(90.0/2)(225.0 + 6.0 + 5.0) - 5.0
 = 158.123 mm

Intermediate Term [Cc]:

= sqrt(2 * pi² * Elastic Modulus / Yield Stress)
 = sqrt(2 * pi² * 199955/239.9)
 = 128.255

Slenderness ratio [KL/r]:

= KL/r
 = 1 * 158.123/32.759
 = 4.827

Bending Moment [Rm]:

= Fl / (2 * Bplen) * e * L / 2
 = 0.0 / (2 * 400.0) * 333.754 * 158.12/2
 = 0.000 N-mm

Compressive Allowable, KL/r < Cc(4.8268 < 128.255) per AISC E2-1 [Sca]:

= (1 - (KLr)² / (2 * Cc²)) Fy / (5/3 + 3 * (KLr) / (8 * Cc) - (KLr³) / (8 * Cc³))
 = (1 - (4.83)² / (2 * 128.25²)) 240 /
 (5/3 + 3 * (4.83) / (8 * 128.25) - (4.83³) / (8 * 128.25³))
 = 142.7 MPa

AISC Unity Check of Outside Ribs (must be <= 1)

= Sc/Sca + (Rm * C1 / I) / Sba
 = 0.71/142.66 + (0.0 * 86.259/1458867) / 159.96
 = 0.005

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 Nozzle Calcs.: Noz i1 Nozl: 27 6:12pm Feb 21, 2022

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Input, Nozzle Desc: Noz i1

From: 10

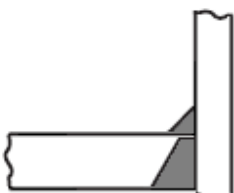
Pressure for Reinforcement Calculations	P	1.4400	MPa
Temperature for Internal Pressure	Temp	150	C
Parent Material		SA/GB 713	
Parent Allowable Stress at Temperature	Sv	145.50	MPa
Parent Allowable Stress At Ambient	Sva	145.50	MPa
Inside Diameter of Elliptical Head	D	450.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	5.0000	mm
Head Internal Corrosion Allowance	c	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		0.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.90	MPa
Allowable Stress At Ambient	Sna	137.90	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		2.9921	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	6.0000	mm
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	10.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	6.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	3.7500	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.

Nozzle Sketch (may not represent actual weld type/configuration)



Insert/Set-in Nozzle No Pad, no Inside projection

Reinforcement CALCULATION, Description: Noz i1

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ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.992 in.
 Actual Thickness Used in Calculation 0.236 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 = $P \cdot D \cdot K1 / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 = $0.9 \cdot 1.44 \cdot 452.0 / (2 \cdot 145.5 \cdot 1.0 - 0.2 \cdot 1.44)$
 = 2.0065 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 = $1.44 \cdot 38.0 / (138 \cdot 1.0 + 0.4 \cdot 1.44)$
 = 0.3952 mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	128.0002	mm
Parallel to Vessel Wall, opening length	d	64.0001	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz i1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation or
 force the Appendix 1-10 computation in Nozzle Design Options.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.3952	mm
Wall Thickness per UG16(b),	tr16b	= 3.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 3.2258	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 3.5000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1, tb2)]
 = min[4.8, max(3.5, 3.5)]
 = 3.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)
 = max(0.3952, 3.5)
 = 3.5000 mm

Available Nozzle Neck Thickness = 6.0000 mm --> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 5.0, tr = 2.007, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot (E^*) / (tg - c) = 0.502$, Temp. Reduction = 32 C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8	C
Min Metal Temp. at Required thickness (UCS 66.1)	-40	C
Min Metal Temp. w/o impact per UG-20(f)	-29	C

Gov. MDMT of the nozzle to shell joint welded assembly : -40 C

PV Elite 23 SP2 Licensee: #Replace this text with your company name and th
 FileName : 20980145-083, 2, 21 _____
 Nozzle Calcs.: Noz i1 Nozl: 27 6:12pm Feb 21, 2022

Weld Size Calculations, Description: Noz i1

Intermediate Calc. for nozzle/shell Welds	Tmin	5.0000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.5000 = 0.7 * t_{min}$	$4.2420 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.7691 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 16.7691 mm

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 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz a1 Nozl: 28 6:12pm Feb 21, 2022

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Input, Nozzle Desc: Noz a1

From: 20

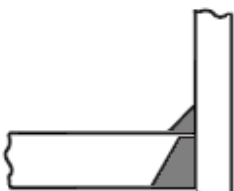
Pressure for Reinforcement Calculations	P	1.4400	MPa
Temperature for Internal Pressure	Temp	150	C
Parent Material		SA/GB 713	
Parent Allowable Stress at Temperature	Sv	145.50	MPa
Parent Allowable Stress At Ambient	Sva	145.50	MPa
Inside Diameter of Cylindrical Shell	D	450.00	mm
Shell Finished (Minimum) Thickness	t	6.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		431.50	mm
User Entered Minimum Design Metal Temperature		0.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.90	MPa
Allowable Stress At Ambient	Sna	117.90	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		0.7430	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	15.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	5.0000	mm
Groove weld depth between Nozzle and Vessel	wgnv	5.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.

Nozzle Sketch (may not represent actual weld type/configuration)



Insert/Set-in Nozzle No Pad, no Inside projection

Reinforcement CALCULATION, Description: Noz a1

PV Elite 23 SP2 Licensee: #Replace this text with your company name and th
 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz a1 Noz: 28 6:12pm Feb 21, 2022

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 0.614 in.
 Actual Thickness Used in Calculation 0.218 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.44 \cdot 226.0 / (146 \cdot 1.0 - 0.6 \cdot 1.44)$
 $= 2.2501$ mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \left(\exp \left(\frac{P}{S \cdot E} \right) - 1 \right)$ per Appendix 1-2 (a)(1)
 $= 8.798 \left(\exp \left(\frac{1.44}{117.9 \cdot 1.0} \right) - 1 \right)$
 $= 0.1081$ mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	36.6700	mm
Parallel to Vessel Wall	Rn+tn+t	18.3350	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	11.3430	mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz a1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation or
 force the Appendix 1-10 computation in Nozzle Design Options.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.1081	mm
Wall Thickness per UG16(b),	tr16b	= 3.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 3.2501	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 3.5000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.5000	mm
Wall Thickness per table UG-45	tb3	= 3.5100	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[3.51, \max(3.5, 3.5)]$
 $= 3.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.1081, 3.5)$
 $= 3.5000$ mm

Available Nozzle Neck Thickness = min3.6 mm --> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 3.6, tr = 0.108, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot (E^*) / (tg - c) = 0.042$, Temp. Reduction = 78 C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29	C
Min Metal Temp. at Required thickness (UCS 66.1)	-104	C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 C

Weld Size Calculations, Description: Noz a1

PV Elite 23 SP2 Licensee: #Replace this text with your company name and th
FileName : 20980145-083, 2, 21
Nozzle Calcs. : Noz a1 Nozl: 28 6:12pm Feb 21, 2022

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Intermediate Calc. for nozzle/shell Welds Tmin 4.5372 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.1760 = 0.7 * t_{min}$	$3.5350 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations.
Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
(small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.3955 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 21.3955 mm

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 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz b Nozl: 29 6:12pm Feb 21, 2022

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Input, Nozzle Desc: Noz b

From: 20

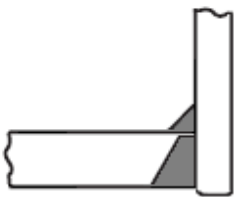
Pressure for Reinforcement Calculations	P	1.4400	MPa
Temperature for Internal Pressure	Temp	150	C
Parent Material		SA/GB 713	
Parent Allowable Stress at Temperature	Sv	145.50	MPa
Parent Allowable Stress At Ambient	Sva	145.50	MPa
Inside Diameter of Cylindrical Shell	D	450.00	mm
Shell Finished (Minimum) Thickness	t	6.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	100.0000	mm
Distance from Bottom/Left Tangent		566.50	mm
User Entered Minimum Design Metal Temperature		0.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.90	MPa
Allowable Stress At Ambient	Sna	137.90	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		115.65	deg
Diameter		1.0630	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	3.5000	mm
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	25.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	5.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	5.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.

Nozzle Sketch (may not represent actual weld type/configuration)



Insert/Set-in Nozzle No Pad, no Inside projection

PV Elite 23 SP2 Licensee: #Replace this text with your company name and th
 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz b Nozl: 29 6:12pm Feb 21, 2022

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: Noz b

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	1.063 in.
Actual Thickness Used in Calculation	0.138 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1.44 \cdot 226.0 / (146 \cdot 1.0 - 0.6 \cdot 1.44)$
 = 2.2501 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 = $1.44 \cdot 13.5 / (138 \cdot 1.0 + 0.4 \cdot 1.44)$
 = 0.1404 mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	DL	49.0465	mm
Parallel to Vessel Wall, opening length	d	24.5232	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	6.2500	mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz b.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation or force the Appendix 1-10 computation in Nozzle Design Options.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 3.5, tr = 0.14, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot (E^*) / (tg - c) = 0.056$, Temp. Reduction = 78 C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 C
Min Metal Temp. w/o impact per UG-20(f)	-29 C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 C

Weld Size Calculations, Description: Noz b

Intermediate Calc. for nozzle/shell welds	Tmin	2.5000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$1.7500 = 0.7 \cdot t_{min}$	$3.5350 = 0.7 \cdot W_o$ mm

*Skipping the nozzle attachment weld strength calculations.
 Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
 (small nozzles) do not require a weld strength check.*

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case	2	MPa
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PV Elite 23 SP2 Licensee: #Replace this text with your company name and th
 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz b NozI: 29 6:12pm Feb 21, 2022

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Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: Noz b

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	1.063 in.
Actual Thickness Used in Calculation	0.138 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1.44 \cdot 226.0 / (146 \cdot 1.0 - 0.6 \cdot 1.44)$
 = 2.2501 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 = $1.44 \cdot 13.5 / (138 \cdot 1.0 + 0.4 \cdot 1.44)$
 = 0.1404 mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	DL	43.9999 mm
Parallel to Vessel Wall, opening length	d	21.9999 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	6.2500 mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz b.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation or force the Appendix 1-10 computation in Nozzle Design Options.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.1404 mm
Wall Thickness per UG16(b),	tr16b = 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 3.2501 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 3.5000 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm
Wall Thickness per table UG-45	tb3 = 3.9464 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[3.946, max(3.5, 3.5)]
 = 3.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(1.1404, 3.5)
 = 3.5000 mm

Available Nozzle Neck Thickness = 3.5000 mm --> OK

Weld Size Calculations, Description: Noz b

Intermediate Calc. for nozzle/shell welds	Tmin	2.5000 mm
---	------	-----------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$1.7500 = 0.7 \cdot t_{min}$	$3.5350 = 0.7 \cdot W_o$ mm

Skipping the nozzle attachment weld strength calculations.

PV Elite 23 SP2 Licensee: #Replace this text with your company name and th
 FileName : 20980145-083,2,21 _____
 Nozzle Calcs.: Noz b Nozl: 29 6:12pm Feb 21, 2022

Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:
 Converged Max. Allow. Pressure in Operating case 2 MPa

The Drop for this Nozzle is : 7.2815 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 38.9581 mm

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 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz d Nozl: 30 6:12pm Feb 21, 2022

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Input, Nozzle Desc: Noz d

From: 20

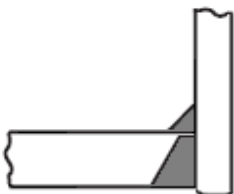
Pressure for Reinforcement Calculations	P	1.4400	MPa
Temperature for Internal Pressure	Temp	150	C
Parent Material		SA/GB 713	
Parent Allowable Stress at Temperature	Sv	145.50	MPa
Parent Allowable Stress At Ambient	Sva	145.50	MPa
Inside Diameter of Cylindrical Shell	D	450.00	mm
Shell Finished (Minimum) Thickness	t	6.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	110.0000	mm
Distance from Bottom/Left Tangent		266.50	mm
User Entered Minimum Design Metal Temperature		0.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.90	MPa
Allowable Stress At Ambient	Sna	137.90	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		118.44	deg
Diameter		1.4961	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	3.5000	mm
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	20.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	5.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	5.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.

Nozzle Sketch (may not represent actual weld type/configuration)



Insert/Set-in Nozzle No Pad, no Inside projection

PV Elite 23 SP2 Licensee: #Replace this text with your company name and th
 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz d Nozl: 30 6:12pm Feb 21, 2022

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: Noz d

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	1.496 in.
Actual Thickness Used in Calculation	0.138 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1.44 \cdot 226.0 / (146 \cdot 1.0 - 0.6 \cdot 1.44)$
 = 2.2501 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 = $1.44 \cdot 19.0 / (138 \cdot 1.0 + 0.4 \cdot 1.44)$
 = 0.1976 mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	DL	75.6053	mm
Parallel to Vessel Wall, opening length	d	37.8027	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	6.2500	mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz d.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation or force the Appendix 1-10 computation in Nozzle Design Options.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 3.5, tr = 0.198, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot (E^*) / (tg - c) = 0.079$, Temp. Reduction = 78 C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 C
Min Metal Temp. w/o impact per UG-20(f)	-29 C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 C

Weld Size Calculations, Description: Noz d

Intermediate Calc. for nozzle/shell welds	Tmin	2.5000	mm
---	------	--------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$1.7500 = 0.7 \cdot t_{min}$	$3.5350 = 0.7 \cdot W_o$ mm

*Skipping the nozzle attachment weld strength calculations.
 Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
 (small nozzles) do not require a weld strength check.*

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case	2	MPa
--	---	-----

PV Elite 23 SP2 Licensee: #Replace this text with your company name and th
 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz d NozI: 30 6:12pm Feb 21, 2022

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Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: Noz d

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	1.496 in.
Actual Thickness Used in Calculation	0.138 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1.44 \cdot 226.0 / (146 \cdot 1.0 - 0.6 \cdot 1.44)$
 = 2.2501 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 = $1.44 \cdot 19.0 / (138 \cdot 1.0 + 0.4 \cdot 1.44)$
 = 0.1976 mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	DL	65.9998	mm
Parallel to Vessel Wall, opening length	d	32.9999	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	6.2500	mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz d.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation or force the Appendix 1-10 computation in Nozzle Design Options.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.1976	mm
Wall Thickness per UG16(b),	tr16b = 3.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 3.2501	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 3.5000	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000	mm
Wall Thickness per table UG-45	tb3 = 4.1200	mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[4.12, max(3.5, 3.5)]
 = 3.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(1.1976, 3.5)
 = 3.5000 mm

Available Nozzle Neck Thickness = 3.5000 mm --> OK

Weld Size Calculations, Description: Noz d

Intermediate Calc. for nozzle/shell welds	Tmin	2.5000	mm
---	------	--------	----

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$1.7500 = 0.7 \cdot t_{min}$	$3.5350 = 0.7 \cdot W_o$ mm

Skipping the nozzle attachment weld strength calculations.

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FileName : 20980145-083, 2, 21 _____
Nozzle Calcs.: Noz d Nozl: 30 6:12pm Feb 21, 2022

Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
(small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:
Converged Max. Allow. Pressure in Operating case 2 MPa

The Drop for this Nozzle is : 11.9303 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 38.7805 mm

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 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz g1 Nozl: 31 6:12pm Feb 21, 2022

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Input, Nozzle Desc: Noz g1

From: 20

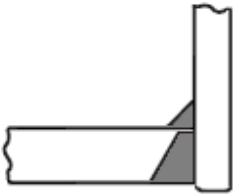
Pressure for Reinforcement Calculations	P	1.4400	MPa
Temperature for Internal Pressure	Temp	150	C
Parent Material		SA/GB 713	
Parent Allowable Stress at Temperature	Sv	145.50	MPa
Parent Allowable Stress At Ambient	Sva	145.50	MPa
Inside Diameter of Cylindrical Shell	D	450.00	mm
Shell Finished (Minimum) Thickness	t	6.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	110.0000	mm
Distance from Bottom/Left Tangent		226.50	mm
User Entered Minimum Design Metal Temperature		0.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.90	MPa
Allowable Stress At Ambient	Sna	137.90	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		61.56	deg
Diameter		0.9843	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	3.5000	mm
Flange Material		SA-182 F304	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	15.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	5.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	5.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Flange Class		2500	
Flange Grade		GR 2.1	

The Pressure Design option was Design Pressure + static head.

Nozzle Sketch (may not represent actual weld type/configuration)



Insert/Set-in Nozzle No Pad, no Inside projection

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: Noz g1

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	0.984 in.
Actual Thickness Used in Calculation	0.138 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.44 \cdot 226.0 / (146 \cdot 1.0 - 0.6 \cdot 1.44)$
 $= 2.2501 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.44 \cdot 12.5 / (138 \cdot 1.0 + 0.4 \cdot 1.44)$
 $= 0.1300 \text{ mm}$

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	45.7570	mm
Parallel to Vessel Wall, opening length	d	22.8785	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	6.2500	mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz g1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation or
 force the Appendix 1-10 computation in Nozzle Design Options.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, Curve: A

Govrn. thk, tg = 3.5, tr = 0.13, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot (E^*) / (tg - c) = 0.052$, Temp. Reduction = 78 C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 C
Min Metal Temp. w/o impact per UG-20(f)	-29 C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 3.5, tr = 0.13, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot (E^*) / (tg - c) = 0.052$, Temp. Reduction = 78 C

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 FileName : 20980145-083, 2, 21
 Nozzle Calcs. : Noz g1 Nozl: 31 6:12pm Feb 21, 2022

Min Metal Temp. w/o impact per UCS-66, Curve A -8 C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 C
 Min Metal Temp. w/o impact per UG-20(f) -29 C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 C

Note:

The flange pressure ratio is less than 0.35. Impact testing is not required due to low stress per UHA-51(g).

Design Pressure/Ambient Rating = $1.44/41.37 = 0.035$

Weld Size Calculations, Description: Noz g1

Intermediate Calc. for nozzle/shell welds Tmin 2.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$1.7500 = 0.7 * t_{min}$	$3.5350 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations.
 Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
 (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case 2 MPa

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: Noz g1

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	0.984 in.
Actual Thickness Used in Calculation	0.138 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.44 \cdot 226.0 / (146 \cdot 1.0 - 0.6 \cdot 1.44)$
 $= 2.2501$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.44 \cdot 12.5 / (138 \cdot 1.0 + 0.4 \cdot 1.44)$
 $= 0.1300$ mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	40.0000 mm
Parallel to Vessel Wall, opening length	d	20.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	6.2500 mm

Note:

Taking a UG-36(c)(3)(a) exemption for nozzle: Noz g1.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation or force the Appendix 1-10 computation in Nozzle Design Options.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

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FileName : 20980145-083, 2, 21
Nozzle Calcs.: Noz g1 Nozl: 31 6:12pm Feb 21, 2022

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Wall Thickness for Internal/External pressures	ta = 1.1300 mm
Wall Thickness per UG16(b),	tr16b = 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 3.2501 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 3.5000 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm
Wall Thickness per table UG-45	tb3 = 3.5100 mm

Determine Nozzle Thickness candidate [tb]:
= min[tb3, max(tb1, tb2)]
= min[3.51, max(3.5, 3.5)]
= 3.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
= max(ta, tb)
= max(1.13, 3.5)
= 3.5000 mm

Available Nozzle Neck Thickness = 3.5000 mm --> OK

Weld Size Calculations, Description: Noz g1

Intermediate Calc. for nozzle/shell welds Tmin 2.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$1.7500 = 0.7 * t_{min}$	$3.5350 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations.
Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
(small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case 2 MPa

The Drop for this Nozzle is : 7.5486 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 29.3987 mm

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 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz x-2 Nozl: 32 6:12pm Feb 21, 2022

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Input, Nozzle Desc: Noz x-2

From: 40

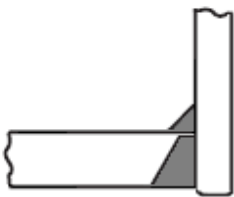
Pressure for Reinforcement Calculations	P	1.4400	MPa
Temperature for Internal Pressure	Temp	150	C
Parent Material		SA/GB 713	
Parent Allowable Stress at Temperature	Sv	142.70	MPa
Parent Allowable Stress At Ambient	Sva	142.70	MPa
Inside Diameter of Cylindrical Shell	D	450.00	mm
Shell Finished (Minimum) Thickness	t	6.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	140.0000	mm
Distance from Bottom/Left Tangent		796.00	mm
User Entered Minimum Design Metal Temperature		0.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.90	MPa
Allowable Stress At Ambient	Sna	117.90	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		52.69	deg
Diameter		0.957	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	30.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	5.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	5.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.

Nozzle Sketch (may not represent actual weld type/configuration)



Insert/Set-in Nozzle No Pad, no Inside projection

PV Elite 23 SP2 Licensee: #Replace this text with your company name and th
FileName : 20980145-083, 2, 21
Nozzle Calcs.: Noz x-2 Nozl: 32 6:12pm Feb 21, 2022

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Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: Noz x-2

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	0.957 in.
Actual Thickness Used in Calculation	0.179 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
= $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.44 \cdot 226.0 / (143 \cdot 1.0 - 0.6 \cdot 1.44)$
= 2.2945 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
= $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
= $1.44 \cdot 17.23 / (118 \cdot 1.0 - 0.6 \cdot 1.44)$
= 0.2120 mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	DL	87.9135	mm
Parallel to Vessel Wall, opening length	d	43.9567	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.6285	mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz x-2.
This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation or force the Appendix 1-10 computation in Nozzle Design Options.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 3.981, tr = 0.212, c = 1.0 mm, E* = 1.0
Thickness Ratio = $tr \cdot (E^*) / (tg - c) = 0.071$, Temp. Reduction = 78 C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 C

Weld Size Calculations, Description: Noz x-2

Intermediate Calc. for nozzle/shell welds	Tmin	3.8514	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.6960 = 0.7 \cdot t_{min}$	$3.5350 = 0.7 \cdot W_o$ mm

*Skipping the nozzle attachment weld strength calculations.
Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
(small nozzles) do not require a weld strength check.*

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: Noz x-2

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	1.278 in.
Actual Thickness Used in Calculation	0.191 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1.44 \cdot 226.0 / (143 \cdot 1.0 - 0.6 \cdot 1.44)$
 = 2.2945 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1.44 \cdot 17.23 / (118 \cdot 1.0 - 0.6 \cdot 1.44)$
 = 0.2120 mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	68.9224 mm
Parallel to Vessel Wall, opening length	d	34.4612 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.6285 mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz x-2.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation or
 force the Appendix 1-10 computation in Nozzle Design Options.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.2120 mm
Wall Thickness per UG16(b),	tr16b	= 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 3.2945 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 3.5000 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.5000 mm
Wall Thickness per table UG-45	tb3	= 4.1200 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[4.12, max(3.5, 3.5)]
 = 3.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(1.212, 3.5)
 = 3.5000 mm

Available Nozzle Neck Thickness = $0.875 \cdot 4.55 = 3.981$ mm --> OK

Weld Size Calculations, Description: Noz x-2

Intermediate Calc. for nozzle/shell welds	Tmin	3.8514 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.6960 = 0.7 \cdot t_{min}$	$3.5350 = 0.7 \cdot W_o$ mm

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FileName : 20980145-083, 2, 21 _____
Nozzle Calcs. : Noz x-2 Nozl: 32 6:12pm Feb 21, 2022

Skipping the nozzle attachment weld strength calculations.
Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
(small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:
Converged Max. Allow. Pressure in Operating case 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 19.0481 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 56.6506 mm

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 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz g2 Nozl: 33 6:12pm Feb 21, 2022

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Input, Nozzle Desc: Noz g2

From: 40

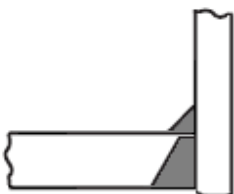
Pressure for Reinforcement Calculations	P	1.4400	MPa
Temperature for Internal Pressure	Temp	150	C
Parent Material		SA/GB 713	
Parent Allowable Stress at Temperature	Sv	142.70	MPa
Parent Allowable Stress At Ambient	Sva	142.70	MPa
Inside Diameter of Cylindrical Shell	D	450.00	mm
Shell Finished (Minimum) Thickness	t	6.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		796.00	mm
User Entered Minimum Design Metal Temperature		0.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.90	MPa
Allowable Stress At Ambient	Sna	137.90	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		90.00	deg
Diameter		0.9843	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	3.5000	mm
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	15.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	5.0000	mm
Groove weld depth between Nozzle and Vessel	wgnv	5.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.

Nozzle Sketch (may not represent actual weld type/configuration)



Insert/Set-in Nozzle No Pad, no Inside projection

Reinforcement CALCULATION, Description: Noz g2

PV Elite 23 SP2 Licensee: #Replace this text with your company name and th
 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz g2 NozI: 33 6:12pm Feb 21, 2022

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 0.984 in.
 Actual Thickness Used in Calculation 0.138 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.44 \cdot 226.0 / (143 \cdot 1.0 - 0.6 \cdot 1.44)$
 $= 2.2945$ mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.44 \cdot 12.5 / (138 \cdot 1.0 + 0.4 \cdot 1.44)$
 $= 0.1300$ mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	40.0000	mm
Parallel to Vessel Wall, opening length	d	20.0000	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	6.2500	mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz g2.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation or
 force the Appendix 1-10 computation in Nozzle Design Options.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.1300	mm
Wall Thickness per UG16(b),	tr16b	= 3.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 3.2945	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 3.5000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.5000	mm
Wall Thickness per table UG-45	tb3	= 3.5100	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[3.51, \max(3.5, 3.5)]$
 $= 3.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.13, 3.5)$
 $= 3.5000$ mm

Available Nozzle Neck Thickness = 3.5000 mm --> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 3.5, tr = 0.13, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot (E^*) / (tg - c) = 0.052$, Temp. Reduction = 78 C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8	C
Min Metal Temp. at Required thickness (UCS 66.1)	-104	C
Min Metal Temp. w/o impact per UG-20(f)	-29	C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 C

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 FileName : 20980145-083, 2, 21 _____
 Nozzle Calcs. : Noz g2 Nozl: 33 6:12pm Feb 21, 2022

Weld Size Calculations, Description: Noz q2

Intermediate Calc. for nozzle/shell welds Tmin 2.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$1.7500 = 0.7 * t_{min}$	$3.5350 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations.
 Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
 (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case 2 MPa

The Drop for this Nozzle is : 0.3475 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 21.3475 mm

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 Nozzle Calcs.: Noz x-1 Nozl: 34 6:12pm Feb 21, 2022

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Input, Nozzle Desc: Noz x-1

From: 40

Pressure for Reinforcement Calculations	P	1.4400	MPa
Temperature for Internal Pressure	Temp	150	C
Parent Material		SA/GB 713	
Parent Allowable Stress at Temperature	Sv	142.70	MPa
Parent Allowable Stress At Ambient	Sva	142.70	MPa
Inside Diameter of Cylindrical Shell	D	450.00	mm
Shell Finished (Minimum) Thickness	t	6.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

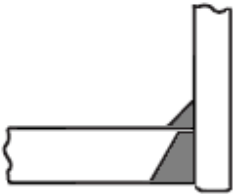
Physical Maximum for Diameter Limit	Dmax	40.0000	mm
Distance from Cylinder/Cone Centerline	L1	80.0000	mm
Distance from Bottom/Left Tangent		796.00	mm
User Entered Minimum Design Metal Temperature		0.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.90	MPa
Allowable Stress At Ambient	Sna	117.90	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		110.26	deg
Diameter		0.957	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	30.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	5.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	5.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.

Nozzle Sketch (may not represent actual weld type/configuration)



Insert/Set-in Nozzle No Pad, no Inside projection

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: Noz x-1

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	0.957 in.
Actual Thickness Used in Calculation	0.179 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.44 \cdot 226.0 / (143 \cdot 1.0 - 0.6 \cdot 1.44)$
 $= 2.2945 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.44 \cdot 17.23 / (118 \cdot 1.0 - 0.6 \cdot 1.44)$
 $= 0.2120 \text{ mm}$

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	42.8054	mm
Parallel to Vessel Wall, opening length	d	21.4027	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.6285	mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz x-1.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation or force the Appendix 1-10 computation in Nozzle Design Options.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, tg = 3.981, tr = 0.212, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot (E^*) / (tg - c) = 0.071$ Temp. Reduction = 78 C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 C

Weld Size Calculations, Description: Noz x-1

Intermediate Calc. for nozzle/shell Welds	Tmin	3.8514	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.6960 = 0.7 * t_{min.}$	$3.5350 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations.
 Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
 (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: Noz x-1

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	0.957 in.
Actual Thickness Used in Calculation	0.179 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.44 \cdot 226.0 / (143 \cdot 1.0 - 0.6 \cdot 1.44)$
 $= 2.2945 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.44 \cdot 17.23 / (118 \cdot 1.0 - 0.6 \cdot 1.44)$
 $= 0.2120 \text{ mm}$

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	40.0000 mm
Parallel to Vessel Wall, opening length	d	20.0000 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	9.6285 mm

Note:

Taking a UG-36(c)(3)(a) exemption for nozzle: Noz x-1.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation or
 force the Appendix 1-10 computation in Nozzle Design Options.

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.2120 mm
Wall Thickness per UG16(b),	tr16b	= 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 3.2945 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 3.5000 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.5000 mm
Wall Thickness per table UG-45	tb3	= 4.1200 mm

Determine Nozzle Thickness candidate [tb]:

$= \min[tb3, \max(tb1, tb2)]$
 $= \min[4.12, \max(3.5, 3.5)]$
 $= 3.5000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(ta, tb)$

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 Nozzle Calcs.: Noz x-1 Nozl: 34 6:12pm Feb 21, 2022

$$= \max(1.212, 3.5)$$

$$= 3.5000 \text{ mm}$$

Available Nozzle Neck Thickness = $0.875 * 4.55 = 3.981 \text{ mm}$ --> OK

Weld Size Calculations, Description: Noz x-1

Intermediate Calc. for nozzle/shell Welds Tmin 3.8514 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.6960 = 0.7 * t_{min}$	$3.5350 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations.
 Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
 (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 9.2814 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 45.6889 mm

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Input, Nozzle Desc: Noz c

From: 40

Pressure for Reinforcement Calculations	P	1.4400	MPa
Temperature for Internal Pressure	Temp	150	C
Parent Material		SA/GB 713	
Parent Allowable Stress at Temperature	Sv	142.70	MPa
Parent Allowable Stress At Ambient	Sva	142.70	MPa
Inside Diameter of Cylindrical Shell	D	450.00	mm
Shell Finished (Minimum) Thickness	t	6.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

Note : User defined Limit(s) of Reinforcement specified below:

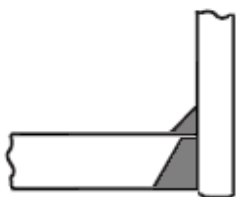
Physical Maximum for Diameter Limit	Dmax	50.0000	mm
Distance from Bottom/Left Tangent		796.00	mm
User Entered Minimum Design Metal Temperature		0.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.90	MPa
Allowable Stress At Ambient	Sna	137.90	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		1.3779	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	3.5000	mm
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	20.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	5.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	5.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
User Defined Nozzle/Shell Centerline Angle		53.0000	deg.

The Pressure Design option was Design Pressure + static head.

Nozzle Sketch (may not represent actual weld type/configuration)



Insert/Set-in Nozzle No Pad, no Inside projection

Reinforcement CALCULATION, Description: Noz c

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	1.378 in.
Actual Thickness Used in Calculation	0.138 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1.44 \cdot 226.0 / (143 \cdot 1.0 - 0.6 \cdot 1.44)$
 = 2.2945 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 = $1.44 \cdot 17.5 / (138 \cdot 1.0 + 0.4 \cdot 1.44)$
 = 0.1820 mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	62.6068 mm
Parallel to Vessel Wall, opening length	d	31.3034 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	6.2500 mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz c.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation or
 force the Appendix 1-10 computation in Nozzle Design Options.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.1820 mm
Wall Thickness per UG16(b),	tr16b	= 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1	= 3.2945 mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 3.5000 mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.5000 mm
Wall Thickness per table UG-45	tb3	= 4.1200 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[4.12, max(3.5, 3.5)]
 = 3.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(1.182, 3.5)
 = 3.5000 mm

Available Nozzle Neck Thickness = 3.5000 mm --> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 3.5, tr = 0.182, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot (E^*) / (tg - c) = 0.073$, Temp. Reduction = 78 C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 C
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 Nozzle Calcs.: Noz c Nozl: 35 6:12pm Feb 21, 2022

Min Metal Temp. at Required thickness (UCS 66.1) -104 C
 Min Metal Temp. w/o impact per UG-20(f) -29 C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 C

Weld Size Calculations, Description: Noz c

Intermediate Calc. for nozzle/shell Welds Tmin 2.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$1.7500 = 0.7 * t_{min}$	$3.5350 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations.
 Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
 (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case 2 MPa

The Drop for this Nozzle is : 0.6816 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 26.6816 mm

Note: This nozzle or its diameter limit intersects weld seam # [1] on this element. The distance between the nozzle and seam is, 0.00 mm , while the distance from the center of the nozzle to the edge of the diameter limit is: 31.30 mm

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Input, Nozzle Desc: Noz h

From: 40

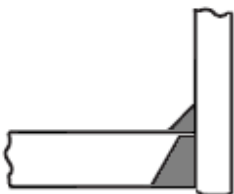
Pressure for Reinforcement Calculations	P	1.4400	MPa
Temperature for Internal Pressure	Temp	150	C
Parent Material		SA/GB 713	
Parent Allowable Stress at Temperature	Sv	142.70	MPa
Parent Allowable Stress At Ambient	Sva	142.70	MPa
Inside Diameter of Cylindrical Shell	D	450.00	mm
Shell Finished (Minimum) Thickness	t	6.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	180.0000	mm
Distance from Bottom/Left Tangent		796.00	mm
User Entered Minimum Design Metal Temperature		0.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.90	MPa
Allowable Stress At Ambient	Sna	137.90	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		141.19	deg
Diameter		1.2598	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	3.5000	mm
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	15.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	5.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	5.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.

Nozzle Sketch (may not represent actual weld type/configuration)



Insert/Set-in Nozzle No Pad, no Inside projection

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 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz h NozI: 36 6:12pm Feb 21, 2022

Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: Noz h

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	1.260 in.
Actual Thickness Used in Calculation	0.138 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1.44 \cdot 226.0 / (143 \cdot 1.0 - 0.6 \cdot 1.44)$
 = 2.2945 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 = $1.44 \cdot 16.0 / (138 \cdot 1.0 + 0.4 \cdot 1.44)$
 = 0.1664 mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	DL	89.4128 mm
Parallel to Vessel Wall, opening length	d	44.7064 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	6.2500 mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz h.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation or force the Appendix 1-10 computation in Nozzle Design Options.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 3.5, tr = 0.166, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot (E^*) / (tg - c) = 0.067$, Temp. Reduction = 78 C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 C
Min Metal Temp. w/o impact per UG-20(f)	-29 C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 C

Weld Size Calculations, Description: Noz h

Intermediate Calc. for nozzle/shell welds	Tmin	2.5000 mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$1.7500 = 0.7 \cdot t_{min}$	$3.5350 = 0.7 \cdot W_o$ mm

*Skipping the nozzle attachment weld strength calculations.
 Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
 (small nozzles) do not require a weld strength check.*

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case	2 MPa
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Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: Noz h

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	1.260 in.
Actual Thickness Used in Calculation	0.138 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.44 \cdot 226.0 / (143 \cdot 1.0 - 0.6 \cdot 1.44)$
 $= 2.2945 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= 1.44 \cdot 16.0 / (138 \cdot 1.0 + 0.4 \cdot 1.44)$
 $= 0.1664 \text{ mm}$

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	DL	53.9999	mm
Parallel to Vessel Wall, opening length	d	26.9999	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	6.2500	mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz h.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation or force the Appendix 1-10 computation in Nozzle Design Options.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.1664	mm
Wall Thickness per UG16(b),	tr16b = 3.5000	mm
Wall Thickness, shell/head, internal pressure	trb1 = 3.2945	mm
Wall Thickness	tb1 = max(trb1, tr16b) = 3.5000	mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000	mm
Wall Thickness per table UG-45	tb3 = 3.9464	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[3.946, \max(3.5, 3.5)]$
 $= 3.5000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.1664, 3.5)$
 $= 3.5000 \text{ mm}$

Available Nozzle Neck Thickness = 3.5000 mm --> OK

Weld Size Calculations, Description: Noz h

Intermediate Calc. for nozzle/shell welds	Tmin	2.5000	mm
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Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$1.7500 = 0.7 \cdot t_{min}$	$3.5350 = 0.7 \cdot W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations.

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Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
(small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case 2 MPa

The Drop for this Nozzle is : 24.5056 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 49.2847 mm

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 Nozzle Calcs.: Noz a2 Nozl: 37 6:12pm Feb 21, 2022

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Input, Nozzle Desc: Noz a2

From: 40

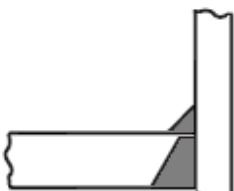
Pressure for Reinforcement Calculations	P	1.4400	MPa
Temperature for Internal Pressure	Temp	150	C
Parent Material		SA/GB 713	
Parent Allowable Stress at Temperature	Sv	142.70	MPa
Parent Allowable Stress At Ambient	Sva	142.70	MPa
Inside Diameter of Cylindrical Shell	D	450.00	mm
Shell Finished (Minimum) Thickness	t	6.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		796.00	mm
User Entered Minimum Design Metal Temperature		0.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	117.90	MPa
Allowable Stress At Ambient	Sna	117.90	MPa
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		0.7430	in.
Size and Thickness Basis		Nominal	
Nominal Thickness		80	
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	20.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	5.0000	mm
Groove weld depth between Nozzle and Vessel	wgnv	5.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.

Nozzle Sketch (may not represent actual weld type/configuration)



Insert/Set-in Nozzle No Pad, no Inside projection

Reinforcement CALCULATION, Description: Noz a2

PV Elite 23 SP2 Licensee: #Replace this text with your company name and th
 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz a2 Noz: 37 6:12pm Feb 21, 2022

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Inside Diameter Used in Calculation 0.614 in.
 Actual Thickness Used in Calculation 0.218 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 1.44 \cdot 226.0 / (143 \cdot 1.0 - 0.6 \cdot 1.44)$
 $= 2.2945$ mm

Reqd thk per App. 1 of Nozzle Wall, trn [Int. Press]
 $= R \left(\exp \left[\frac{P}{S \cdot E} \right] - 1 \right)$ per Appendix 1-2 (a)(1)
 $= 8.798 \left(\exp \left[\frac{1.44}{(117.9 \cdot 1.0)} - 1 \right] \right)$
 $= 0.1081$ mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	36.6700	mm
Parallel to Vessel Wall	Rn+tn+t	18.3350	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	11.3430	mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz a2.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation or
 force the Appendix 1-10 computation in Nozzle Design Options.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 1.1081	mm
Wall Thickness per UG16(b),	tr16b	= 3.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 3.2945	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 3.5000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.5000	mm
Wall Thickness per table UG-45	tb3	= 3.5100	mm

Determine Nozzle Thickness candidate [tb]:
 $= \min[tb3, \max(tb1, tb2)]$
 $= \min[3.51, \max(3.5, 3.5)]$
 $= 3.5000$ mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 $= \max(ta, tb)$
 $= \max(1.1081, 3.5)$
 $= 3.5000$ mm

Available Nozzle Neck Thickness = min3.6 mm --> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, tg = 3.6, tr = 0.108, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot (E^*) / (tg - c) = 0.042$, Temp. Reduction = 78 C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29	C
Min Metal Temp. at Required thickness (UCS 66.1)	-104	C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 C

Weld Size Calculations, Description: Noz a2

PV Elite 23 SP2 Licensee: #Replace this text with your company name and th
 FileName : 20980145-083, 2, 21 _____
 Nozzle Calcs.: Noz a2 Nozl: 37 6:12pm Feb 21, 2022

Intermediate Calc. for nozzle/shell Welds T_{min} 4.5372 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.1760 = 0.7 * t_{min}$	$3.5350 = 0.7 * W_o$ mm

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 0.3955 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 26.3955 mm

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 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz f Nozl: 38 6:12pm Feb 21, 2022

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Input, Nozzle Desc: Noz f

From: 40

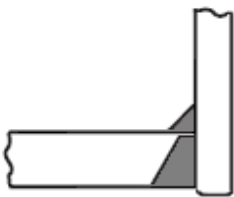
Pressure for Reinforcement Calculations	P	1.4400	MPa
Temperature for Internal Pressure	Temp	150	C
Parent Material		SA/GB 713	
Parent Allowable Stress at Temperature	Sv	142.70	MPa
Parent Allowable Stress At Ambient	Sva	142.70	MPa
Inside Diameter of Cylindrical Shell	D	450.00	mm
Shell Finished (Minimum) Thickness	t	6.0000	mm
Shell Internal Corrosion Allowance	c	1.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Cylinder/Cone Centerline	L1	105.0000	mm
Distance from Bottom/Left Tangent		796.00	mm
User Entered Minimum Design Metal Temperature		0.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.90	MPa
Allowable Stress At Ambient	Sna	137.90	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		-27.04	deg
Diameter		1.4961	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	3.5000	mm
Corrosion Allowance	can	1.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	20.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	5.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	5.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.

Nozzle Sketch (may not represent actual weld type/configuration)



Insert/Set-in Nozzle No Pad, no Inside projection

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 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz f Nozl: 38 6:12pm Feb 21, 2022

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Note : Checking Nozzle 90 degrees to the Longitudinal axis.

Reinforcement CALCULATION, Description: Noz f

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	1.496 in.
Actual Thickness Used in Calculation	0.138 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1.44 \cdot 226.0 / (143 \cdot 1.0 - 0.6 \cdot 1.44)$
 = 2.2945 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 = $1.44 \cdot 19.0 / (138 \cdot 1.0 + 0.4 \cdot 1.44)$
 = 0.1976 mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	DL	74.5814 mm
Parallel to Vessel Wall, opening length	d	37.2907 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	6.2500 mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz f.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation or force the Appendix 1-10 computation in Nozzle Design Options.*

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 3.5, tr = 0.198, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot (E^*) / (tg - c) = 0.079$, Temp. Reduction = 78 C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8 C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 C
Min Metal Temp. w/o impact per UG-20(f)	-29 C

Gov. MDMT of the nozzle to shell joint welded assembly : -104 C

Weld Size Calculations, Description: Noz f

Intermediate Calc. for nozzle/shell welds	Tmin	2.5000 mm
---	------	-----------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$1.7500 = 0.7 \cdot t_{min}$	$3.5350 = 0.7 \cdot W_o$ mm

Skipping the nozzle attachment weld strength calculations.
 Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
 (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case	2 MPa
--	-------

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 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz f NozI: 38 6:12pm Feb 21, 2022

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Note : Checking Nozzle in plane parallel to the vessel axis.

Reinforcement CALCULATION, Description: Noz f

ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Outside Diameter Used in Calculation	1.496 in.
Actual Thickness Used in Calculation	0.138 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 = $P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 = $1.44 \cdot 226.0 / (143 \cdot 1.0 - 0.6 \cdot 1.44)$
 = 2.2945 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 = $1.44 \cdot 19.0 / (138 \cdot 1.0 + 0.4 \cdot 1.44)$
 = 0.1976 mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	DL	65.9998 mm
Parallel to Vessel Wall, opening length	d	32.9999 mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	6.2500 mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz f.
 This calculation is valid for nozzles that meet all the requirements of paragraph UG-36. Please check the Code carefully, especially for nozzles that are not isolated or do not meet Code spacing requirements. To force the computation of areas for small nozzles go to Tools->Configuration and check the box to force the UG-37 small nozzle area calculation or force the Appendix 1-10 computation in Nozzle Design Options.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta = 1.1976 mm
Wall Thickness per UG16(b),	tr16b = 3.5000 mm
Wall Thickness, shell/head, internal pressure	trb1 = 3.2945 mm
Wall Thickness	tb1 = max(trb1, tr16b) = 3.5000 mm
Wall Thickness	tb2 = max(trb2, tr16b) = 3.5000 mm
Wall Thickness per table UG-45	tb3 = 4.1200 mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[4.12, max(3.5, 3.5)]
 = 3.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(1.1976, 3.5)
 = 3.5000 mm

Available Nozzle Neck Thickness = 3.5000 mm --> OK

Weld Size Calculations, Description: Noz f

Intermediate Calc. for nozzle/shell welds	Tmin	2.5000 mm
---	------	-----------

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$1.7500 = 0.7 \cdot t_{min}$	$3.5350 = 0.7 \cdot W_o$ mm

Skipping the nozzle attachment weld strength calculations.

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FileName : 20980145-083, 2, 21 _____
Nozzle Calcs.: Noz f Nozl: 38 6:12pm Feb 21, 2022

Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a)
(small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:
Converged Max. Allow. Pressure in Operating case 2 MPa

The Drop for this Nozzle is : 11.2503 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 38.0099 mm

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 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz i2 Nozl: 39 6:12pm Feb 21, 2022

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Input, Nozzle Desc: Noz i2

From: 50

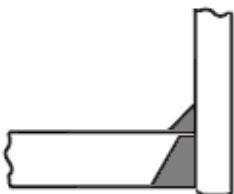
Pressure for Reinforcement Calculations	P	1.4400	MPa
Temperature for Internal Pressure	Temp	150	C
Parent Material		SA/GB 713	
Parent Allowable Stress at Temperature	Sv	145.50	MPa
Parent Allowable Stress At Ambient	Sva	145.50	MPa
Inside Diameter of Elliptical Head	D	450.00	mm
Aspect Ratio of Elliptical Head	Ar	2.00	
Head Finished (Minimum) Thickness	t	5.0000	mm
Head Internal Corrosion Allowance	c	1.0000	mm
Head External Corrosion Allowance	co	0.0000	mm
Distance from Head Centerline	L1	0.0000	mm
User Entered Minimum Design Metal Temperature		0.00	C

Type of Element Connected to the Parent : Nozzle

Material		SA-105	
Material UNS Number		K03504	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	137.90	MPa
Allowable Stress At Ambient	Sna	137.90	MPa
Diameter Basis (for tr calc only)		Outside	
Layout Angle		0.00	deg
Diameter		2.9921	in.
Size and Thickness Basis		Actual	
Actual Thickness	tn	6.0000	mm
Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	10.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	6.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	3.7500	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	wi	0.0000	mm

The Pressure Design option was Design Pressure + static head.

Nozzle Sketch (may not represent actual weld type/configuration)



Insert/Set-in Nozzle No Pad, no Inside projection

Reinforcement CALCULATION, Description: Noz i2

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 FileName : 20980145-083, 2, 21
 Nozzle Calcs.: Noz i2 NozI: 39 6:12pm Feb 21, 2022

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ASME Code, Section VIII, Div. 1, 2019, UG-37 to UG-45

Actual Outside Diameter Used in Calculation 2.992 in.
 Actual Thickness Used in Calculation 0.236 in.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Elliptical Head, tr [Int. Press]
 = $P \cdot D \cdot K_1 / (2 \cdot S_v \cdot E - 0.2 \cdot P)$ per Appendix 1-4(c)
 = $0.9 \cdot 1.44 \cdot 452.0 / (2 \cdot 145.5 \cdot 1.0 - 0.2 \cdot 1.44)$
 = 2.0065 mm

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 = $P \cdot R_o / (S_n \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 = $1.44 \cdot 38.0 / (138 \cdot 1.0 + 0.4 \cdot 1.44)$
 = 0.3952 mm

UG-40, Limits of Reinforcement : [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	128.0002	mm
Parallel to Vessel Wall, opening length	d	64.0001	mm
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	10.0000	mm

Note:

*Taking a UG-36(c)(3)(a) exemption for nozzle: Noz i2.
 This calculation is valid for nozzles that meet all the requirements of
 paragraph UG-36. Please check the Code carefully, especially for nozzles
 that are not isolated or do not meet Code spacing requirements. To force
 the computation of areas for small nozzles go to Tools->Configuration
 and check the box to force the UG-37 small nozzle area calculation or
 force the Appendix 1-10 computation in Nozzle Design Options.*

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	ta	= 0.3952	mm
Wall Thickness per UG16(b),	tr16b	= 3.5000	mm
Wall Thickness, shell/head, internal pressure	trb1	= 3.2258	mm
Wall Thickness	tb1 = max(trb1, tr16b)	= 3.5000	mm
Wall Thickness	tb2 = max(trb2, tr16b)	= 3.5000	mm
Wall Thickness per table UG-45	tb3	= 4.8000	mm

Determine Nozzle Thickness candidate [tb]:
 = min[tb3, max(tb1, tb2)]
 = min[4.8, max(3.5, 3.5)]
 = 3.5000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:
 = max(ta, tb)
 = max(0.3952, 3.5)
 = 3.5000 mm

Available Nozzle Neck Thickness = 6.0000 mm --> OK

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), min(Curve:A, Curve:B)

Govrn. thk, tg = 5.0, tr = 2.007, c = 1.0 mm, E* = 1.0
 Thickness Ratio = $tr \cdot (E^*) / (tg - c) = 0.502$, Temp. Reduction = 32 C

Min Metal Temp. w/o impact per UCS-66, Curve A	-8	C
Min Metal Temp. at Required thickness (UCS 66.1)	-40	C
Min Metal Temp. w/o impact per UG-20(f)	-29	C

Gov. MDMT of the nozzle to shell joint welded assembly : -40 C

Intermediate Calc. for nozzle/shell welds Tmin 5.0000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$3.5000 = 0.7 * t_{min}$	$4.2420 = 0.7 * W_o \text{ mm}$

Skipping the nozzle attachment weld strength calculations. Per UW-15(b)(2) the nozzles exempted by UG-36(c)(3)(a) (small nozzles) do not require a weld strength check.

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Max. Allow. Pressure in Operating case 2 MPa

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 1.7691 mm
The Cut Length for this Nozzle is, Drop + Ho + H + T : 16.7691 mm

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 FileName : SQ-FL
 Flange Analysis : Flange x-1 Item: 8 6:12pm Feb 21,2022

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Input Echo, Flange Item 8, Description: Flange x-1

Description of Flange Geometry (Type)		Loose Ring	
Description of Flange Analysis	Partial,	Thickness	
Print Final Results for Given Thickness		Yes	
Design Pressure	P	1.44	MPa
Design Temperature		150.00	C
Corrosion Allowance	ci	1.0000	mm
Flange Inside Diameter	B	30.000	mm
Flange Outside Diameter	A	164.000	mm
Flange Thickness	t	18.0000	mm
Perform thickness calcs. based on rigidity		Yes	
Flange Material (Not Normalized)		SA/GB 713	
Flange Allowable Stress At Temperature	Sfo	143	MPa
Flange Allowable Stress At Ambient	Sfa	143	MPa
Bolt Material		SA-193 B7	
Bolt Material UNS Number		G41400	
Bolt Allowable Stress At Temperature	Sb	172	MPa
Bolt Allowable Stress At Ambient	Sa	172	MPa
Length of Weld Leg at Back of Ring	tw	6.0000	mm
Number of Splits in Ring Flange	nsp	0	
Diameter of Bolt Circle	C	119.000	mm
Nominal Bolt Diameter	a	16.0000	mm
Type of Thread Series		TEMA Metric Thread	
Number of Bolts	n	4	
Compute Full Flange Design Bolt Load (S*ab) ?		No	
Flange Face Outside Diameter	Fod	116.000	mm
Flange Face Inside Diameter	Fid	85.000	mm
Flange Facing Sketch		Code Sketch 1a	
Gasket Outside Diameter	Go	116.000	mm
Gasket Inside Diameter	Gi	85.000	mm
Gasket Factor,	m	2.0000	
Gasket Design Seating Stress	y	14.700	MPa
Column for Gasket Seating	2,	Code Column II	
Gasket Thickness	tg	3.0000	mm
Shell Material		SA-106 B	
Shell Material UNS Number		K03006	
Shell Allowable Stress At Temperature	So	118	MPa
Shell Allowable Stress At Ambient	Sa	118	MPa
Full face Gasket Flange Option		Program Selects	

ASME Code, Section VIII, Division 1, 2019

Flange analysis, Flange number 8, Description: Flange x-1
ASME Code, Section VIII, Div. 1, 2019 App. 2

Code R Dimension,	$R = (C-B)/2 - g_1$	40.500 mm
Gasket Contact Width,	$N = (Goc-Gic) / 2$	15.5 mm
Basic Gasket Width,	$b_0 = N / 2.0$	7.75 mm
Effective Gasket Width,	$b = \text{SQRT}(b_0) * 2.5$	6.96 mm
Gasket Reaction Diameter,	$G = (Go+ Gi)/2$	100.5 mm

PV Elite 23 SP2 Licensee: #Replace this text with your company name and th
 FileName : SQ-FL _____
 Flange Analysis : Flange x-1 Item: 8 6:12pm Feb 21, 2022

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure[H]:

$$= 0.785 * G * G * P_{eq}$$

$$= 0.785 * 100.5 * 100.5 * 1.4400$$

$$= 11417.3226 \text{ N}$$

Contact Load on Gasket Surfaces[Hp]:

$$= 2 * b * \pi * G * m * P$$

$$= 2 * 6.96 * 3.1416 * 100.5 * 2.0000 * 1.44$$

$$= 12656.956 \text{ N}$$

Hydrostatic End Load at Flange ID[Hd]:

$$= 0.785 * B_{cor} * B_{cor} * P$$

$$= 0.785 * 38.0000 * 38.0000 * 1.4400$$

$$= 1633.13 \text{ N}$$

Pressure Force on Flange Face[Ht]:

$$= H - H_d$$

$$= 11417.3226 - 1633.13$$

$$= 9784.197103 \text{ N}$$

Operating Bolt Load[Wm1]:

$$= H + H_p + H'_p \text{ (cannot be } < 0 \text{)}$$

$$= (11417.3226 + 12656.956 + 0)$$

$$= 24074.2786 \text{ N}$$

Gasket Seating Bolt Load[Wm2]:

$$= y * b * \pi * G + y_{Part} * b_{Part} * l_p$$

$$= 14.70 * 6.96 * 3.141 * 100.5 + .00 * .0000 * .00$$

$$= 32285.23147 \text{ N}$$

Required Bolt Area[Am]:

$$= \text{Maximum of } W_{m1}/S_b, W_{m2}/S_a$$

$$= \text{Maximum of } 24074.2786 / 172, 32285.23147 / 172$$

$$= 187.7048342 \text{ mm}^2$$

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:

$$= 2a + 6t / (m + 0.5)$$

$$= 2 * 16.000 + 6 * 17.000 / (2.00 + 0.5)$$

$$= 72.8 \text{ mm} \text{ **Maximum Bolt Spacing Exceeded !**}$$

For vessels in lethal service or when specified by the user/agent, the maximum bolt spacing shall not exceed the value of [Bsmax]. See ASME VIII-1 Appendix 2 paragraph 2-5(d) for more information.

Actual Circumferential Bolt Spacing [Bs]:

$$= C * \sin(\pi / n)$$

$$= 119.000 * \sin(3.142 / 4)$$

$$= 84.11 \text{ mm}$$

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:

$$= \max(\sqrt{Bs / (2a + t)}, 1)$$

$$= \max(\sqrt{84.11 / (2 * 16.000 + 17.000)}, 1)$$

$$= 1.31$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

Distance Across Corners for Nuts		24.250	mm
Circular Wrench End Diameter	a	38.000	mm
	Minimum	Actual	Maximum
Bolt Area, mm ²	187.7	660.185	
Radial Distance between Hub and Bolts:	15.400	38.000	

FileName : SQ-FL

Flange Analysis : Flange x-1 Item: 8 6:12pm Feb 21, 2022

Radial Distance between Bolts and Edge: 15.400 23.000
 Circ. Spacing between the Bolts: 31.600 84.11 72.8

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calculation]:

$$= A_b * S_b / (y * \pi * (G_o + G_i))$$

$$= 660.185 * 172 / (14.70 * 3.14 * (116.00 + 85.00))$$

$$= 12.24 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating[W]:

$$= S_a * (A_m + A_b) / 2.0$$

$$= 172 * (187.7 + 660.185) / 2.0$$

$$= 72918.52574 \text{ N}$$

Gasket Seating Force[Hg]:

$$= W_{m1} - H$$

$$= 24074.2786 - 11417.3226$$

$$= 12656.956 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction[hg]:

$$= (C - G) / 2.0$$

$$= (119.0000 - 100.5) / 2.0$$

$$= 9.25 \text{ mm}$$

Distance to Face Pressure Reaction[ht]:

$$= (h_d + h_g) / 2.0$$

$$= (40.5000 + 9.250) / 2.0$$

$$= 24.875 \text{ mm}$$

Distance to End Pressure Reaction[hd]:

$$= (C - B_{cor}) / 2.0$$

$$= (119.0000 - 38.0000) / 2.0$$

$$= 40.5000 \text{ mm}$$

Summary of Moments for Internal Pressure:

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1633.13	40.5	1.31	86657.45N-mm
Face Pressure, Mt	9784.20	24.875	1.31	318874.46N-mm
Gasket Load, Mg	12656.96	9.25	1.31	153391.91N-mm
Gasket Seating, Ma	72918.53	9.25	1.31	883712.63N-mm

Total Moment for Operation, Mo	558923.8384N-mm
Total Moment for Gasket Seating, Ma	883712.625N-mm

Effective Hub Length, ho =	0.000 mm
Hub Ratio, h/ho = Defined as 0.0	0.000
Thickness Ratio, g1/g0 = Defined as 0.0	0.000

Factors from Figure 2-7.1	K =	4.21
T = 0.977	U =	1.524
Y = 1.39	Z =	1.12

Tangential Flange Stress, Operating[Sto]:

$$= (Y * M_o) / (t^2 * B_{cor})$$

$$= (1.39 * 558923.8384) / (17.0000^2 * 38.0000)$$

$$= 70.56 \text{ MPa}$$

Tangential Flange Stress, Seating[Sta]:

$$= (Y * M_a) / (t^2 * B_{cor})$$

$$= (1.39 * 883712.625) / (17.0000^2 * 38.0000)$$

$$= 111.56 \text{ MPa}$$

FileName : SQ-FL _____

Flange Analysis : Flange x-1 Item: 8 6:12pm Feb 21, 2022

Bolt Stress, Operating[Bso]:

$$= (Wm1 / Ab)$$

$$= (24074.2786 / 660.185)$$

$$= 36.47 \text{ MPa}$$

Bolt Stress, Seating[Bsa]:

$$= (Wm2 / Ab)$$

$$= (32285.23147 / 660.185)$$

$$= 48.9 \text{ MPa}$$

Stress Computation Results:

	Actual	Operating Allowed	Gasket Actual	Seating Allowed
Tangential Flange	70.56	143.	111.56	143. MPa
Bolting	36.47	172.	48.9	172. MPa

Minimum Required Flange Thickness + CA	16.5 mm
Estimated M.A.W.P. (Operating)	2.34 MPa
Estimated M.A.W.P. (Gasket Seating)	2.47 MPa
Estimated Finished Weight of Flange	27.9 N
Estimated Unfinished Weight of Forging	27.9 N

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:

$$= 109.4 * Ma / Bsc * Cnv_fac / (Eamb * t^3 * \ln(K) * Kl)$$

$$= 109.4 * 883712.625 / 1.31 * 1.000 / (202713 * 17.000^3$$

$$* 1.438 * 0.20)$$

$$= 0.34 \text{ (should be } \leq 1)$$

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:

$$= 109.4 * Mo / Bsc * Cnv_fac / (Eop * tc^3 * \ln(K) * Kl)$$

$$= 109.4 * 558923.8384 / 1.31 * 1.000 / (195074 * 17.000^3$$

$$* 1.438 * 0.20)$$

$$= 0.224 \text{ (should be } \leq 1)$$

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 FileName : SQ-FL _____
 Flange Analysis : Flange x-2 Item: 9 6:12pm Feb 21, 2022

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Input Echo, Flange Item 9, Description: Flange x-2

Description of Flange Geometry (Type)		Loose Ring	
Description of Flange Analysis	Partial,	Thickness	
Print Final Results for Given Thickness		Yes	
Design Pressure	P	1.44	MPa
Design Temperature		150.00	C
Corrosion Allowance	ci	1.0000	mm
Flange Inside Diameter	B	30.000	mm
Flange Outside Diameter	A	115.000	mm
Flange Thickness	t	18.0000	mm
Perform thickness calcs. based on rigidity		Yes	
Flange Material (Not Normalized)		SA/GB 713	
Flange Allowable Stress At Temperature	Sfo	143	MPa
Flange Allowable Stress At Ambient	Sfa	143	MPa
Bolt Material		SA-193 B7	
Bolt Material UNS Number		G41400	
Bolt Allowable Stress At Temperature	Sb	172	MPa
Bolt Allowable Stress At Ambient	Sa	172	MPa
Length of Weld Leg at Back of Ring	tw	6.0000	mm
Number of Splits in Ring Flange	nsp	0	
Diameter of Bolt Circle	C	60.000	mm
Nominal Bolt Diameter	a	8.0000	mm
Type of Thread Series		TEMA Metric Thread	
Number of Bolts	n	4	
Compute Full Flange Design Bolt Load (S*ab) ?		No	
Flange Face Outside Diameter	Fod	82.000	mm
Flange Face Inside Diameter	Fid	31.000	mm
Flange Facing Sketch		Code Sketch 1a	
Gasket Outside Diameter	Go	82.000	mm
Gasket Inside Diameter	Gi	31.000	mm
Gasket Factor,	m	2.0000	
Gasket Design Seating Stress	y	14.700	MPa
Column for Gasket Seating	2,	Code Column II	
Gasket Thickness	tg	3.0000	mm
Shell Material		SA-106 B	
Shell Material UNS Number		K03006	
Shell Allowable Stress At Temperature	So	118	MPa
Shell Allowable Stress At Ambient	Sa	118	MPa
Full face Gasket Flange Option		Program Selects	

ASME Code, Section VIII, Division 1, 2019

Flange analysis, Flange number 9, Description: Flange x-2
ASME Code, Section VIII, Div. 1, 2019 App. 2

Code R Dimension,	$R = (C-B)/2 - g1$	14.000 mm
Basic Gasket Width,	$b0 = N / 2.0$	11.75 mm
Effective Gasket Width,	$b = \text{SQRT}(b0) * 2.5$	8.57 mm
Gasket Reaction Diameter,	$G = (Go + Gi)/2$	58.5 mm

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Flange Analysis : Flange x-2 Item: 9 6:12pm Feb 21, 2022

Gasket Reaction Diameter, $G = C - 2.0 * hg$ 47.040 mm

Note: This gasket extends beyond the Bolt circle diameter, it will be designed per Taylor Forge Bulletin NO. 45.

Basic Flange and Bolt loads:

Hydrostatic End Load due to Pressure[H]:
= $0.785 * G * G * Peq$
= $0.785 * 58.5 * 58.5 * 1.4400$
= 3868.5114 N

Contact Load on Gasket Surfaces[Hp]:
= $2 * b * PI * G * m * P$
= $2 * 8.57 * 3.1416 * 58.5 * 2.0000 * 1.44$
= 9071.668188 N

Hydrostatic End Load at Flange ID[Hd]:
= $0.785 * Bcor * Bcor * P$
= $0.785 * 32.0000 * 32.0000 * 1.4400$
= 1158.116696 N

Pressure Force on Flange Face[Ht]:
= $H - Hd$
= $3868.5114 - 1158.116696$
= 2710.394704N

Operating Bolt Load[Wm1]:
= $H + Hp + H'p$ (cannot be < 0)
= $(3868.5114 + 39071.668188 + 0)$
= 12940.17959 N

Gasket Seating Bolt Load[Wm2]:
= $y * b * PI * G + yPart * bPart * lp$
= $14.70 * 8.57 * 3.141 * 58.500 + .00 * .0000 * .00$
= 23139.91668 N

Required Bolt Area[Am]:
= Maximum of Wm1/Sb, Wm2/Sa
= Maximum of $12940.17959 / 172$, $23139.91668 / 172$
= 134.5343993 mm²

ASME Maximum Circumferential Spacing between Bolts per App. 2 eq. (3) [Bsmax]:
= $2a + 6t / (m + 0.5)$
= $2 * 8.000 + 6 * 17.000 / (2.00 + 0.5)$
= 56.8mm

Actual Circumferential Bolt Spacing [Bs]:
= $C * \sin(\pi / n)$
= $60.000 * \sin(3.142 / 4)$
= 42.4 mm

ASME Moment Multiplier for Bolt Spacing per App. 2 eq. (7) [Bsc]:
= $\max(\sqrt{ Bs / (2a + t) }, 1)$
= $\max(\sqrt{ 42.4 / (2 * 8.000 + 17.000) }, 1)$
= 1.13

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

Distance Across Corners for Nuts		24.250	mm
Circular Wrench End Diameter	a	38.000	mm

	Minimum	Actual	Maximum

Bolt Area, mm ²	134.53	145.19	
Radial Distance between Hub and Bolts:	15.400	9.000	

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FileName : SQ-FL

Flange Analysis : Flange x-2 Item: 9 6:12pm Feb 21, 2022

Radial Distance between Bolts and Edge: 15.400 27.500
Circ. Spacing between the Bolts: 31.600 42.4 56.800

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calculation]:

$$= Ab * Sb / (y * PI * (Go + Gi))$$

$$= 145.19 * 172 / (14.70 * 3.14 * (82.00 + 31.00))$$

$$= 4.62 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating[W]:

$$= Sa * (Am + Ab) / 2.0$$

$$= 172 * (134.53 + 145.19) / 2.0$$

$$= 24056.60794 \text{ N}$$

Gasket Seating Force[Hg]:

$$= Wm1 - H$$

$$= 12940.17959 - 3868.5114$$

$$= 9071.668188 \text{ N}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction[hg]:

$$= (C - G) / 2.0$$

$$= (60.0000 - 58.5000) / 2.0$$

$$= 0.75 \text{ mm}$$

Distance to Face Pressure Reaction[ht]:

$$= (hd + hg) / 2.0$$

$$= (14.000 + 0.75000) / 2.0$$

$$= 7.37500 \text{ mm}$$

Distance to End Pressure Reaction[hd]:

$$= (C - Bcor) / 2.0$$

$$= (60.0000 - 32.0000) / 2.0$$

$$= 14.000 \text{ mm}$$

Summary of Moments for Internal Pressure:

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	1158.1	14	1.13	18380.40261 N-mm
Face Pressure, Mt	2710.4	7.375	1.13	22660.48634 N-mm
Gasket Load, Mg	9071.67	0.75	1.13	7712.995571 N-mm
Gasket Seating, Matm	24056.61	0.75	1.13	20453.62624 N-mm
Total Moment for Operation, Mo				48753.88453 N-mm
Total Moment for Gasket Seating, Ma				20453.62624 N-mm

Effective Hub Length, ho = 0.000 mm

Hub Ratio, h/ho = Defined as 0.0 0.000

Thickness Ratio, g1/g0 = Defined as 0.0 0.000

Factors from Figure 2-7.1 K = 3.5

T = 1.098 U = 1.78

Y = 1.62 Z = 1.18

Tangential Flange Stress, Operating[Sto]:

$$= (Y * Mo) / (t^2 * Bcor)$$

$$= (1.62 * 48753.88453) / (17.0000^2 * 32.0000)$$

$$= 8.55 \text{ MPa}$$

Tangential Flange Stress, Seating[Sta]:

$$= (Y * Ma) / (t^2 * Bcor)$$

$$= (1.62 * 20453.62624) / (17.0000^2 * 32.0000)$$

$$= 3.59 \text{ MPa}$$

Bolt Stress, Operating[Bso]:

$$= (Wm1 / Ab)$$

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 Flange Analysis : Flange x-2 Item: 9 6:12pm Feb 21, 2022

$$= (12940.17959 / 145.19)$$

$$= 89.12 \text{MPa}$$

Bolt Stress, Seating[Bsa]:

$$= (Wm2 / Ab)$$

$$= (23139.91668 / 145.19)$$

$$= 159.37 \text{ MPa}$$

Stress Computation Results:

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Tangential Flange	8.55	143.	3.59	143. MPa
Bolting	89.12	172.	159.37	172. MPa

Minimum Required Flange Thickness + CA	15 mm
Estimated M.A.W.P. (Operating)	10.94 MPa
Estimated M.A.W.P. (Gasket Seating)	10.18 MPa
Estimated Finished Weight of Flange	5.9 N
Estimated Unfinished Weight of Forging	5.9 N

Flange Rigidity Index, Seating (rotation check) per APP. 2 [Js]:

$$= 109.4 * Ma / Bsc * Cnv_fac / (Eamb * t^3 * \ln(K) * Kl)$$

$$= 109.4 * 20453.62624 / 1.13 * 1.000 / (202713 * 17.000^3$$

$$* 1.25 * 0.20)$$

$$= 0.02 \text{ (should be } \leq 1)$$

Flange Rigidity Index Operating (rotation check) per APP. 2 [J]:

$$= 109.4 * Mo / Bsc * Cnv_fac / (Eop * tc^3 * \ln(K) * Kl)$$

$$= 109.4 * 48753.88453 / 1.13 * 1.000 / (195074 * 17.000^3$$

$$* 1.25 * 0.20)$$

$$= 0.056 \text{ (should be } \leq 1)$$

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